

The mitochondrial stress-induced protein carboxyl-terminal alanine and threonine tailing (msiCAT-tailing) promotes glioblastoma tumorigenesis by modulating mitochondrial functions

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Abstract

The rapid and sustained proliferation in cancer cells requires accelerated protein synthesis. Accelerated protein synthesis and disordered cell metabolism in cancer cells greatly increase the risk of translation errors. ribosome-associated quality control (RQC) is a recently discovered mechanism for resolving ribosome collisions caused by frequent translation stalls. The role of the RQC pathway in cancer initiation and progression remains controversial and confusing. In this study, we investigated the pathogenic role of mitochondrial stress-induced protein carboxyl-terminal terminal alanine and threonine tailing (msiCAT-tailing) in glioblastoma (GBM), which is a specific RQC response to translational arrest on the outer mitochondrial membrane. We found that msiCAT-tailed mitochondrial proteins frequently exist in glioblastoma stem cells (GSCs). Ectopically expressed msiCAT-tailed mitochondrial ATP synthase F1 subunit alpha (ATP5 α) protein increases the mitochondrial membrane potential and blocks mitochondrial permeability transition pore (MPTP) formation/opening. These changes in mitochondrial properties confer resistance to staurosporine (STS)-induced apoptosis in GBM cells. Therefore, msiCAT-tailing can promote cell survival and migration, while genetic and pharmacological inhibition of msiCAT-tailing can prevent the overgrowth of GBM cells.

Highlights

- The RQC pathway is disturbed in glioblastoma (GBM) cells
- msiCAT-tailing on ATP5 α elevates mitochondrial membrane potential and inhibits MPTP opening
- msiCAT-tailing on ATP5 α inhibits drug-induced apoptosis in GBM cells
- Inhibition of msiCAT-tailing impedes overall growth of GBM cells

eLife assessment

Glioblastoma is one of the most aggressive cancers without a cure. Glioblastoma cells are known to have high mitochondrial potential. This **useful** study demonstrates the critical role of the ribosome-associated quality control (RQC) pathway in regulating mitochondrial membrane potential and glioblastoma growth. Some assays are **incomplete**; further revision will improve the significance of this study.

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Introduction

Proteins are the molecular basis of life activities. Excessive protein synthesis is particularly important in rapidly dividing cells, such as cancer cells. To cope with this demand, cancer cells extensively regulate the initiation, elongation and termination of their protein translation (1). However, increased protein translation also proportionally accumulates the likelihood of translation errors (2). Combined with the significant changes in cellular metabolism, such as energy fluctuations and redox state shifts, how to respond to various sudden abnormal events during the protein translation process becomes essential. Ribosome-associated quality control (RQC) is a recently discovered class of mechanisms in eukaryotic cells that sense and rescue ribosome decelerations, stalls, and even collisions that occur during translation elongation or termination steps (3, 4).

RQC consists of a sequence of molecular events. The first is that the ZNF598/RACK1 complex recognizes the characteristic 40S-40S interface formed by collided ribosomes and promotes the ubiquitination of specific 40S subunit proteins (5, 6). Subsequently, the ASC-1 complex dissociates the leading collided ribosome (7, 8). Events that occur thereafter include: ribosomal subunit splitting and recycling (9), modification of the nascent peptide chains through the C-terminal alanine and threonine (CAT-tailing) mechanism (10), release of CAT-tailed products from the 60S subunits by ANKZF1/VMS1 (11), and degradation of aberrant peptides by the Ltn1/VCP/NEMF complex (4). The biological significance of the CAT-tailed translation products generated in the RQC remains unclear. On one side of the coin, CAT-tails can push lysine residues out of the ribosome exit channel to achieve Ltn1-mediated ubiquitination (12), or promote degradation of faulty nascent peptide chains (13, 14); on the other side, CAT-tailed proteins are prone to form detergent-insoluble aggregates (15, 16). In addition, depending on the nature and subcellular localization of the original target, CAT-tailed proteins may also have unique functions, but this is currently poorly understood. Importantly, CAT-tailed proteins have been found in a variety of human neurodegenerative diseases and are believed to play important roles in their pathogenesis (17–19).

The increased abnormalities in translation have been found in cancer cells, such as stop codon readthrough (20), frame-shifting (21), and oxidative stress-induced ribosome stalling (22), indicating the possible involvement of the RQC pathway. However, despite the detection of CAT-tail modification on mitochondrial proteins caused by impaired RQC in HeLa cells (17), there is still a lack of mechanistic studies on the role of RQC factors in cancer biology. The expression of many RQC factors (*e.g.*, ASCC3, ABCE1, ANKZF1 and VCP) has been shown to be dysregulated in cancer cells (23–26). Intriguingly, different RQC factors, or even the same factor, may exhibit completely opposite functions in oncogenesis and tumor suppression under distinct conditions. Inhibition of ABCE1, ASCC3 and VCP inhibits cancer cell growth and survival (23, 24, 26), whereas inhibition of NEMF/Cln and ZNF598 may promote cancer growth and survival (27,

(28 [↗](#)). This suggests that the role of RQC factors in cancer cells is sophisticated and highly genetic and environmental context dependent. A recent study investigated the mechanism of ANKZF1 in mitochondrial proteostasis and its impact on the malignant progression of GBM (29 [↗](#)). However, the nonphysiologically mitochondria-targeted GFP was used in the study to induce the matrix proteotoxicity, and the role of endogenous mitochondrial proteins in this process remains unclear.

The co-translational import defects triggered by mitochondrial stress induce CAT-tailing not only on the mitochondrial C-I30 protein (complex-I 30 kD subunit protein, also known as NDUF53), but also widely on other nuclear genome-encoded mitochondrial proteins (17 [↗](#), 30 [↗](#)). How these mitochondrial stress-induced CAT-tailed (msiCAT-tail) proteins participate in and influence the core of mitochondrial biology has not yet been studied. Since CAT-tailing imparts new properties to target proteins, it is likely to contribute to some unique characteristics of cancer cell mitochondria, such as mitochondrial hyperpolarization (31 [↗](#), 32 [↗](#)) and high mitochondrial membrane potential-related resistance to drug-induced apoptosis (33 [↗](#)–35 [↗](#)). Mitochondrial membrane potential is formed by the proton (H^+) concentration gradient across the mitochondrial inner membrane. It is generated and maintained by oxidative phosphorylation (OXPHOS), in which H^+ ions are shuttled through the electron transport chain composed of four complexes (I to IV), resulting in the accumulation of H^+ in the intermembrane space (36 [↗](#)). ATP synthase (complex-V) then harnesses the energy from the H^+ gradient by promoting its influx into the mitochondrial matrix and produces ATPs in the process (37 [↗](#)). Despite increased metabolic demands, many cancer cells have reduced OXPHOS (38 [↗](#)). It remains unclear how cancer cells maintain or even increase mitochondrial membrane potential in the absence of OXPHOS (31 [↗](#)).

In this study, we focused on msiCAT-tailing modification on the mitochondrial ATP synthase F1 subunit alpha (ATP5 α). We observed that msiCAT-tailed ATP5 α is present in GBM. The short-tailed ATP5 α can incorporate into the ATP synthase, increase mitochondrial membrane potential, and inhibit the assembly and opening of mitochondrial permeability transition pore (MPTP). Therefore, the presence of msiCAT-tailed ATP5 α enhances the resistance of GBM cells to staurosporine (STS)- and temozolomide (TMZ)-induced programmed cell death and promotes cancer cell survival, proliferation, and migration. Conversely, inhibition of msiCAT-tailing suppresses cancer cell growth and resensitizes GBM cells to apoptosis. Our study elucidates the role of CAT-tailed mitochondrial proteins in cancer cells, reveals the importance of the RQC pathway in cancer biology, and indicates that components and products in the RQC pathway may be effective therapeutic targets for GBM.

Results

Presence of mitochondrial stress-induced protein carboxyl-terminal terminal alanine and threonine tailing (msiCAT-tailing) in glioblastoma cells

Dysregulation of individual factors associated with ribosome-associated quality control (RQC) has been described in a variety of cancer cells such as prostate cancer, adenocarcinoma, non-small cell lung cancer and colon cancer cells (23 [↗](#)–26 [↗](#)), but an overall characterization of the RQC pathway genes has not been performed. We first performed differential expression analysis using the transcriptomic data between 153 glioblastoma (GBM) patients and 206 healthy controls from public datasets (39 [↗](#)). We found that compared with normal subjects, the expression levels of genes related with the RQC pathway such as *ABCE1*, *ASCC1*, *ASCC2*, *ASCC3*, *RACK1*, and *VCP* were significantly elevated ($\log FC > 1$; $\text{adj.P.Val} < 0.001$) in GBM cells, whereas *ANKZF1* was the only gene whose expression level ($\log FC = -0.43$, $\text{adj.P.Val} = 0.0005$) was downregulated (Fig. 1A [↗](#), Table S1 [↗](#)). Changes in these genes imply activation of the RQC pathway and accumulation of CAT-tailed protein. The msiCAT-tailed mitochondrial protein C-I 30 (complex-I 30 kD subunit, also

known as NDUF53) was previously discovered in HeLa (cervical cancer) cells as the endogenous substrate of RQC (17 [↗](#)). We wanted to know whether C-I 30 and other msiCAT-tailed proteins are indeed present in GBM cells. Patient-derived GBM stem cells (pGSCs), but not normal neural stem cells (NSCs), displayed several msiCAT-tailed mitochondrial proteins, such as C-I30, COX4 (cytochrome c oxidase subunit 4), and ATP5 α (ATP synthase F1 subunit alpha). In line with the detectable msiCAT-tailed proteins in pGSCs, we also found elevated levels of NEMF (nuclear export mediator factor) (10 [↗](#)) but decreased expression of ANKZF1 (ankyrin repeat and zinc-finger peptidyl tRNA hydrolase 1) (11 [↗](#)), indicating an increase in CAT-tailing activation, which is similar to the results of bioinformatics analysis in GBMs (Fig. 1B [↗](#)). In the following experiments, we selected two GBM cell lines, SF268 (labeled as SF cells in figures) (40 [↗](#)) and GSC827 (41 [↗](#)) with more stem cell characteristics, and two control cell lines SVG p12 (abbreviated as SVG) and Normal Human Astrocytes E6/E7/hTERT (abbreviated as NHA) (42 [↗](#)). We also examined their expression of RQC proteins (Fig. S1A [↗](#)). Similar to the results with patient-derived GSC cells, we also observed decreased ANKZF1 expression and increased ASCC3 in GSC827 and SF268 cells. Although there was no obvious change in NEMF expression, we found modification on NEMF (indicated by *) in GSC827 cells, hinting its possible functional changes (Fig. S1A [↗](#)). In line with this, we used non-stop protein translation system to induce CAT-tailing on Flag tagged β -globin reporter in GBM and control cells (43 [↗](#)). We are surprised to observe that the translation rate of GBM cells for normal proteins (β -globin-control) actually decreased compared to control cells, the RQC system in GBM cells could produce significantly more CAT-tailed protein (β -globin-nonstop); and this process can be inhibited by the CAT-tailing elongation inhibitor anisomycin (44 [↗](#)) (Fig. 1C [↗](#)).

We next wanted to know the biological consequences of generating these CAT-tailed mitochondrial proteins. The heterogeneity of CAT-tailing toward substrates and tail lengths makes it difficult to study the function of specific CAT-tail-modified proteins. In our previous study, adding the AT repeat tails to the C-terminus of mitochondrial proteins can effectively simulate a CAT-tail like reaction (17 [↗](#)). Among the CAT-tailed mitochondrial targets we found, ATP5 α is one of those with the highest abundance in functional mitochondria (45 [↗](#)) and has additional critical roles in cancers (46 [↗](#)). We therefore wondered what unique functions CAT-tailed ATP5 α might provide. To test it, in control (SVG and NHA) and GBM (SF268 and GSC827) cell lines, we overexpressed ATP5 α proteins having artificial CAT-tails: ATP5 α -AT3 containing three alanine-threonine repeats and ATP5 α -AT20 that contains containing twenty alanine-threonine repeats. Consistent with previous observations, long CAT-tailed but not short CAT-tailed proteins, has post-translational modifications and detergent-insoluble aggregates that appear as slower migrating bands and high molecular weight smeared signals in protein electrophoresis (Fig. 1D [↗](#)). Notably, compared with the control cells (SVG, NHA), the GBM cell lines (GSC827, SF268) expressed more ATP5 α -AT20 proteins, indicating that GBM cells have more capability to deal with protein aggregation (Fig. 1D [↗](#), ratios). In addition, cytological analysis tells us that the short AT tail does not significantly affect subcellular localization. Like the tailless ATP5 α protein, ATP5 α -AT3 is mainly located in mitochondria, while a significant proportion of ATP5 α -AT20 appears in the cytoplasm near mitochondria and forms protein aggregates (Fig. S1B [↗](#)). Moreover, the proportion of cells having protein aggregates was highest in GSC cells, which are the most malignant (Fig. S1C [↗](#)). Importantly, not only exogenously expressed tailed proteins, endogenous ATP5 α also formed protein aggregates attached to mitochondria (outer membrane) in some GBM cells (Fig. 1E [↗](#), 1F [↗](#)). Taken together, we found evidence showing the existence of RQC dysregulation and msiCAT-tailing occurring in GBM cells.

msiCAT-tailed ATP5 α elevates mitochondrial membrane potential ($\Delta\Psi_m$)

Cancer cells can keep their mitochondrial membrane potential ($\Delta\Psi_m$) stable, sometimes even elevated, while their respiration is reduced. We first confirmed this phenomenon in patient-derived pGSC cells, which had a higher membrane potential but lower ATP production than

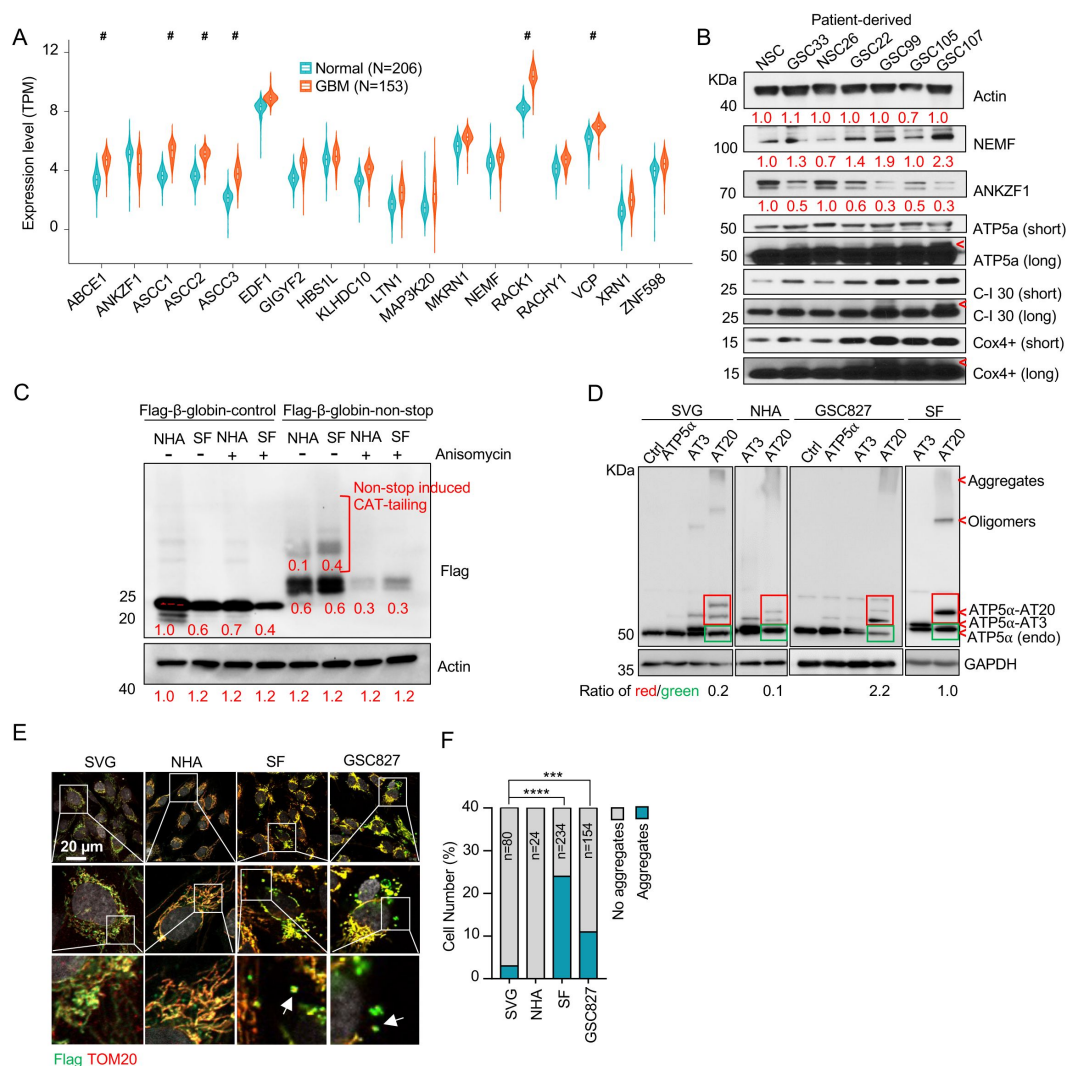


Figure 1.

Evidence for msiCAT-tailing on nuclear-encoded mitochondrial proteins in GBM cells

A. The expression levels of RQC genes changed significantly in GBM tumor tissues (N=153) and normal brain tissues (N=206). Significance was calculated by using unpaired student's t-test, #, logFC (fold change) > 1, adj.P.Val < 0.001.

B. Western blot analysis of msiCAT-tailed mitochondrial proteins and RQC factors in invasive patient-derived GSC and control NSC cells. Actin serves as the loading control. Red arrow heads indicate short CAT-tailed mitochondrial proteins. "short" and "long" represent exposure time. Red numbers represent fold changes in protein levels compared to controls (here NSC), labeling of fold changes in the figures below follows the same rule.

C. Western blot analysis of 5xFLAG-tagged β-globin reporter proteins in GBM and control cells, showing that the RQC system of GBM cells generates more CAT-tailed proteins. Actin serves as the loading control. Red numbers represent fold changes in protein levels compared to the control (left lane: NHA, no anisomycin).

D. Western blot analysis of overexpressed ATP5α-AT3 and ATP5α-AT20 in GBM and control cells. GAPDH serves as the loading control. SVG, SVG p12 cells; NHA, normal human astrocytes; SF, SF268 cells; GSC, GSC827 cells; labeling of cell lines in the figures below follows the same rule. Ctrl, empty vector; ATP5α, non-AT-tailed protein; AT3, ATP5α-AT3; AT20, ATP5α-AT20; labeling of transgenes in the figures below follows the same rule. Arrowheads from bottom to top indicate endogenous ATP5α, ATP5α-AT3, ATP5α-AT20, oligomers and aggregates of msiCAT-tailed ATP5α proteins.

E. Immunofluorescence staining shows that endogenous ATP5α proteins can form aggregates in GBM cells. TOM20 (red) serves as the mitochondrial marker. Arrowheads in GBM cells indicate aggregates composed by ATP5α. In this and subsequent figures, the scale bar is indicated in the images.

F. Quantification of G. n=3, chi-squared test. ***, $P < 0.001$; ****, $P < 0.0001$. All data are representative of at least 3 independent experiments. The total number of cells counted is indicated in each column.

control NSC cells (**Fig. 2A**, **2B**). We also tested our GBM cell lines (GSC827, SF268) and the control cell line NHA (42). We found that both GSC and SF268 exhibited comparable or higher membrane potential than NHA cells (**Fig. S2A**), while their mitochondrial ATP levels were lower (**Fig. S2B**, **S2C**). Inhibiting msiCAT-tailing using genetic approaches, such as knocking down NEMF (sgNEMF) and overexpressing ANZKF1 (oeANZKF1), effectively reduced the mitochondrial membrane potential in GBM but not NHA cells (**Fig. 2C**, **S2D**).

We next wanted to determine how msiCAT-tailed proteins significantly impact mitochondrial functions. We ectopically expressed ATP5 α -AT3 and ATP5 α -AT20 (Flag tagged) proteins in GBM and control cell lines. We found that they both can increase mitochondrial membrane potential in GBM but not NHA cells (**Fig. 2D**). Intriguingly, Blue Native Polyacrylamide Gel Electrophoresis (BN-PAGE) analysis indicated that CAT tails of different lengths exert distinct effects on mitochondria. ATP5 α -AT3 proteins were incorporated into mitochondrial respiratory chain complexes, while ATP5 α -AT20 proteins formed high molecular weight complexes or existed in monomers (**Fig. 2E**). As expected, long CAT-tails form protein aggregates, which may cause proteostasis stress in mitochondria (17, 29). We next utilized the Agilent Seahorse Cell Mitochondrial Stress Test to measure key parameters of mitochondrial respiration by directly measuring the oxygen consumption rate (OCR) of cells. Interestingly, we found that the expression of both ATP5 α -AT3 and ATP5 α -AT20 can impair mitochondrial oxidative phosphorylation, resulting in decreased ATP synthesis, basal respiration, and maximum respiration rates (**Fig. 2F**-**2I**). Collectively, these data indicated that expression of CAT-tailed ATP5 α protein significantly affects mitochondrial function. In comparison, short tails are more likely to directly influence the function of ATP synthase, whereas long tails may cause mitochondrial proteostasis stress thereby indirectly affecting mitochondrial respiration.

msiCAT-tailed ATP5 α regulates mitochondrial permeability transition pore (MPTP) opening

In addition to its primary function of generating ATP, recent studies have implicated F₁F₀ ATP synthase may be an integral component of mitochondrial permeability transition pore (MPTP) complex (47-49). If MPTP is affected by CAT-tailed components such as ATP5 α , this may cause dysregulation of its open-close state or assembly. Therefore, in the following experiments, we investigated how msiCAT-tailing modulates the status of MPTP. By comparing the MPTP status in GBM cells and control cells, we found that MPTP in GSC827 cells is largely in a close state. Interestingly, the CAT-tailing inhibitor anisomycin can effectively eliminate the endogenous ATP5 α aggregation (**Fig. S3A**, **S3B**) and promote the opening of MPTP in GSC827 cells, illustrated by the reduced Calcein signal (**Fig. 3A**, **3B**). Short-term opening of MPTP promote the efflux of proton, Ca²⁺ ions and other signaling molecules from the mitochondrial matrix (50). Mitochondrial Ca²⁺ retention capacity (CRC) measurements have been used to calculate the amount of Ca²⁺ ions needed to open MPTP, directly indicating their open-close states. We observed that GSC827 cells establish higher CRC than NHA cells, and anisomycin pre-treatment and NEMF knockdown significantly reduce the CRC, suggesting the opening of MPTP in GBM cells upon elimination of CAT-tailed proteins (**Fig. 3C**, **3D**).

To determine whether the CAT-tail modifications on ATP5 α can modulate the open-close state of MPTP, we compared the effects of artificial AT repeat tails on MPTP in GBM cells. Interestingly, we found that the short AT tail (AT3) can inhibit the functional opening of MPTP, while the long AT tail (AT20) has a weaker effect (**Fig. 3E**, **3F**), which may be explained by their differential involvement in ATP synthase (**Fig. 2E**). To gain a glimpse into the role of ATP5 α protein with different msiCAT-tails, we examined the interaction between ATP5 α and two known MPTP components Cyp-D and ANT2; however, no direct interaction was detected (**Fig. S3C**). Interestingly, we found that Cyp-D expression is reduced in varying proportions, when ATP5 α -AT3 or ATP5 α -AT20 is ectopically expressed, hinting reduced MPTP formation (**Fig. S3D**). We then used BN-PAGE to detect the complex composed of ANT1/2. We found that the presence of both

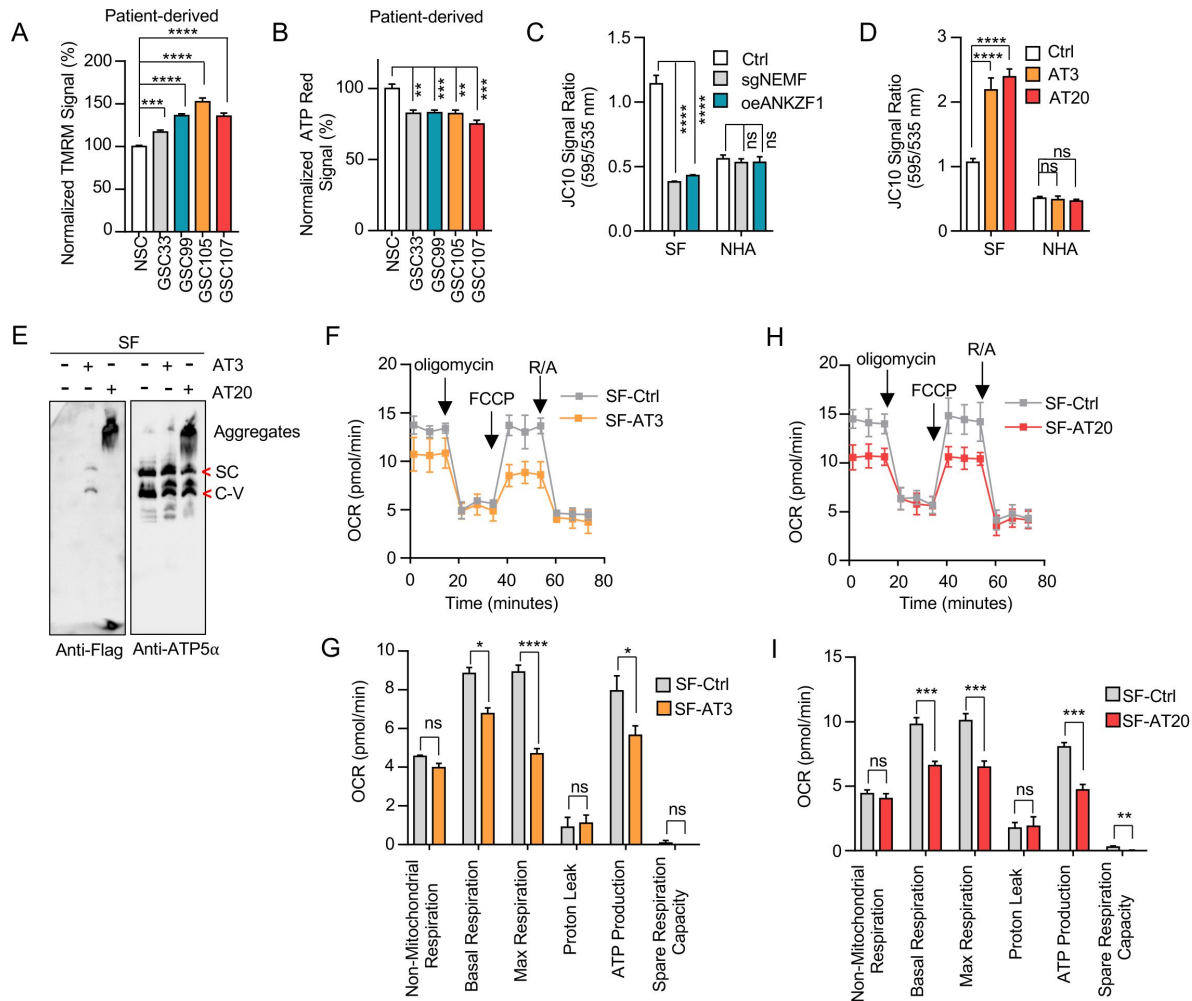


Figure 2.

msiCAT-tailed ATP5α proteins impair mitochondrial respiration but increase mitochondrial membrane potential in GBM cells

A. TMRM staining shows that patient-derived GSC (pGSC) cells display high mitochondrial membrane potential. $n=3$, unpaired student's t-test. ***, $P < 0.001$; ****, $P < 0.0001$.

B. ATP measurement shows that patient-derived GSC (pGSC) cells has low mitochondrial ATP production. $n=3$, unpaired student's t-test. **, $P < 0.01$; ***, $P < 0.001$.

C. JC-10 staining shows that genetic inhibition of msiCAT-tailing pathway reduces mitochondrial membrane potential in GBM but not in NHA control cells. $n=3$, unpaired student's t-test. ****, $P < 0.0001$; ns, not significant.

D. JC-10 staining shows that ectopic expression of ATP5α-AT3 and ATP5α-AT20 increases mitochondrial membrane potential in GBM but not in control cells. $n=3$, unpaired student's t-test. ****, $P < 0.0001$; ns, not significant.

E. BN-PAGE western blot for detecting ATP synthase assembly shows that ATP5α-AT3 is incorporated into the composition of complex V, but ATP5α-AT20 forms high molecular weight protein aggregates in mitochondria of GBM cells. SC: respiratory supercomplex; C-V: complex V/ATP synthase.

F, H. OCR indicates that the reduction of mitochondrial consumption rate of SF268 cells with ATP5α-AT3 and ATP5α-AT20 expression. Oligomycin (1.5 μM), FCCP (1.0 μM), and rotenone/antimycin A (R/A, 0.5 μM) were added sequentially to the reaction cells.

G, I. Statistical summary of mitochondrial respiration parameters in F and H, including non-mitochondrial respiration, basal respiration, maximum respiration, spare respiration, proton leaks and ATP production. $n=3$, unpaired student's t-test. *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$; ****, $P < 0.0001$; ns, not significant.

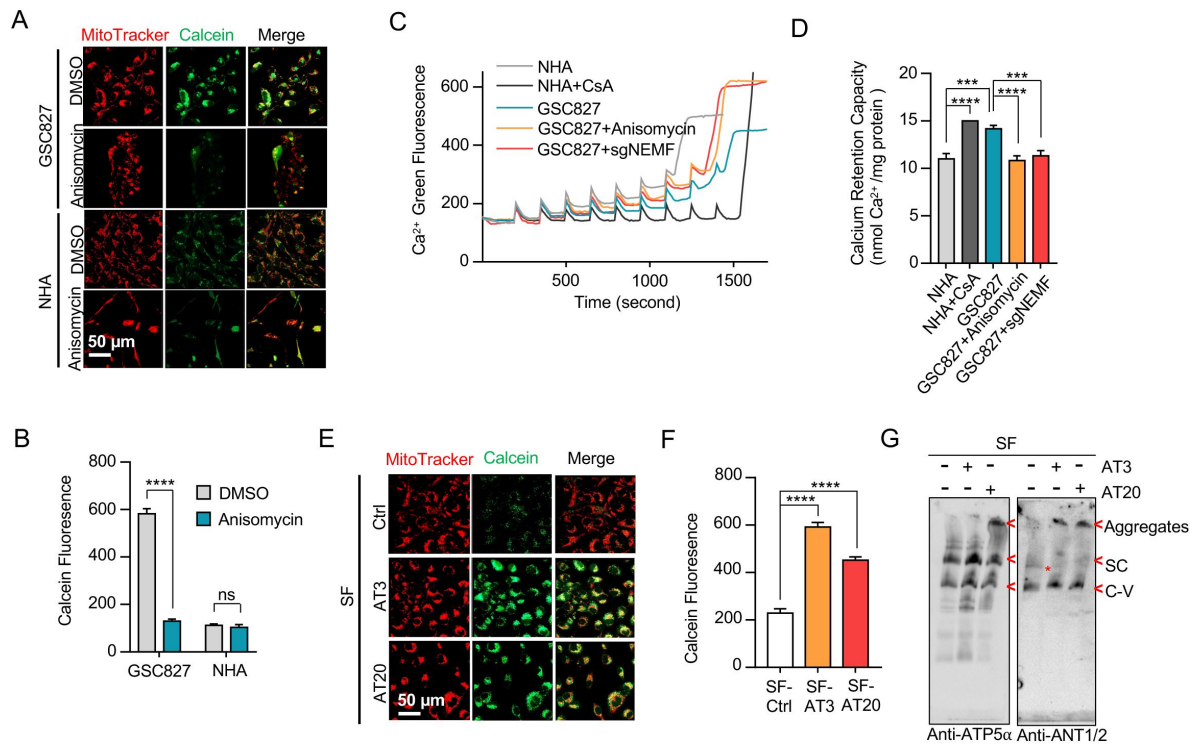


Figure 3.

msiCAT-tailing product regulates mPTP status in GBM cells

- A. Mitochondrial transition pore assay shows that GBM has reduced mPTP opening compared to NHA controls, which can be reversed by transient anisomycin treatment (200 nM).
- B. Quantification of A. $n=3$, unpaired student's t-test. ****, $P < 0.0001$.
- C. Calcium retention capacity (CRC) of isolated mitochondria from cells measured by using Calcium Green-5N dye. NHA is used as the control cell line. Cyclosporin A (CsA, a mPTP inhibitor)-treated sample serves as a positive control.
- D. Statistic of C. The data shows that GBM cells exhibited high CRC, whereas mitochondria from cells pretreated with anisomycin (200 nM) or with NEMF knockout (sgNEMF) are attenuated. $n=3$, unpaired student's t-test. ***, $P < 0.001$; ****, $P < 0.0001$.
- E. Mitochondrial transition pore assay shows that ectopic expression of ATP5 α -AT3 and ATP5 α -AT20 inhibits mPTP opening in GBM cells.
- F. Quantification of E. $n=3$, unpaired student's t-test. ****, $P < 0.0001$.
- G. BN-PAGE western blot shows that the expression of ATP5 α -AT3 and ATP5 α -AT20 changes the pattern of ANT1/2 protein in GBM cells, missing a band (marked by a red * on the blot), and forming high molecular weight protein aggregates. SC: respiratory supercomplex; C-V: complex V/ATP synthase.

ATP5 α -AT3 and -AT20 significantly changed the pattern of ANT1/2-containing complexes, with the expected bands disappearing (indicated by *) and aggregates forming at high-molecular weight positions (**Fig. 3G**). It is worth mentioning that the ANT1/2 complex is in close proximity to ATP synthase in the electrophoresis, supporting the hypothesis that ATP synthase may be part of the MPTP supercomplex. In summary, we conclude that msiCAT-tailed ATP5 α proteins, especially the short-tailed (AT3) ones, incorporate into the ATP synthase and regulate the MPTP status.

msiCAT-tailing promotes GBM cells migration and enhances their resistance to apoptosis

The high mitochondrial membrane potential and tightened MPTP conferred by msiCAT-tailed ATP5 α and other mitochondrial proteins may make cells more adaptable to stresses. Therefore, we next wanted to know how the msiCAT-tailing mechanism affects GBM cells at the cellular level. In the MTT assays (**Fig. 4A**), we observed that overexpression of the short AT repeat tail (AT3) and the long AT repeat tail (AT20) fused with ATP5 α significantly promotes GBM cell but not NHA cell viability (**Fig. 4A**, **4B**). Further, in *in vitro* transwell cell migration assays (**Fig. 4C**) and wound healing experiments (**Fig. 4D**), we observed that GBM cells overexpressing CAT-tailed ATP5 α showed stronger cell invasion ability and faster wound healing, indicating faster cell migration occurs (**Fig. 4C**, **4D**, **S4A**, **S4B**). It is worth mentioning that in ANKZF1 knockdown U87 and U251 cell lines, the accumulation of abnormal mitoGFP can lead to a decrease in cellular adaptability (**Fig. 29**). We speculate that this is because the mitochondria of different cell lines have different adaptability to proteostasis stress. One support was that in the initial experiments, we did not find that the mild expression of ATP5 α -AT3 and ATP5 α -AT20 induces strong mitochondrial proteotoxic responses, such as significantly up-regulated *LONP1*, *mtHSP70* and *HSP60* mRNA level (**Fig. S4C**).

In addition to this, we found that GBM cells are more resistant to staurosporine (STS)-induced apoptosis, as evidenced by reduced TUNEL-positive cells (**Fig. S4D**, **S4E**) and significantly weakened cleavage of Poly (ADP-ribose) polymerase (PARP-1) proteins (**Fig. S4F**) (**S4G**). To confirm whether CAT-tailed ATP5 α proteins contribute to resistance to apoptosis, we further overexpressed these proteins in GBM cells. We found that both short tail (ATP5 α -AT3) and long tail (ATP5 α -AT20) can effectively enhance cells resistance to STS-induced apoptosis, proving that protein CAT-tailation caused may be closely related to tumorigenesis (**Fig. 4E**, **4F**). Consistent with this, we found that overexpression of artificial CAT-tailed ATP5 α proteins, also enhance the resistance of GBM cells to temozolomide (TMZ)-induced apoptosis (**Fig. 4G**). Combining these data, we demonstrate that RQC-induced CAT-tailing on ATP5 α protein may contribute to the resistance of GBM cells to drug-induced apoptosis.

Inhibiting the RQC pathway impedes GBM cell progression

Through previous studies, we confirmed the importance of the RQC pathway mediated msiCAT-tailing mechanism mediated for GBM progression. Therefore, inhibiting the RQC pathway may be a potentially effective means to combat GBM. To test this hypothesis, we first treated patient-derived pGSC cells with the CAT-tailing inhibitor anisomycin. We found that the pGSC cell lines were more sensitive to anisomycin treatment than control NSC cells (**Fig. 5A**). Similarly, by knocking down NEMF (sgNEMF) or overexpressing ANKZF1 (oeANKZF1) in GBM cell line (SF268), we found that genetically inhibiting the RQC pathway can also inhibit GBM growth (**Fig. 5B**). Notably, the growth of NHA cells as a control was also inhibited, indicating that these two genes are critical factors in general cell proliferation (**Fig. S5A**). The RQC pathway has a more significant impact on the migration ability of GBM cells. In transwell *in vitro* cell migration assays, sgNEMF or oeANKZF1 significantly inhibited the migration of GBM cells but had no effect on NHA cells (**Fig. 5C**, **5D**). Similarly, GSC cells but not NHA cells treated with anisomycin also significantly weakened their cell migration ability (**Fig. S5B**, **S5C**).

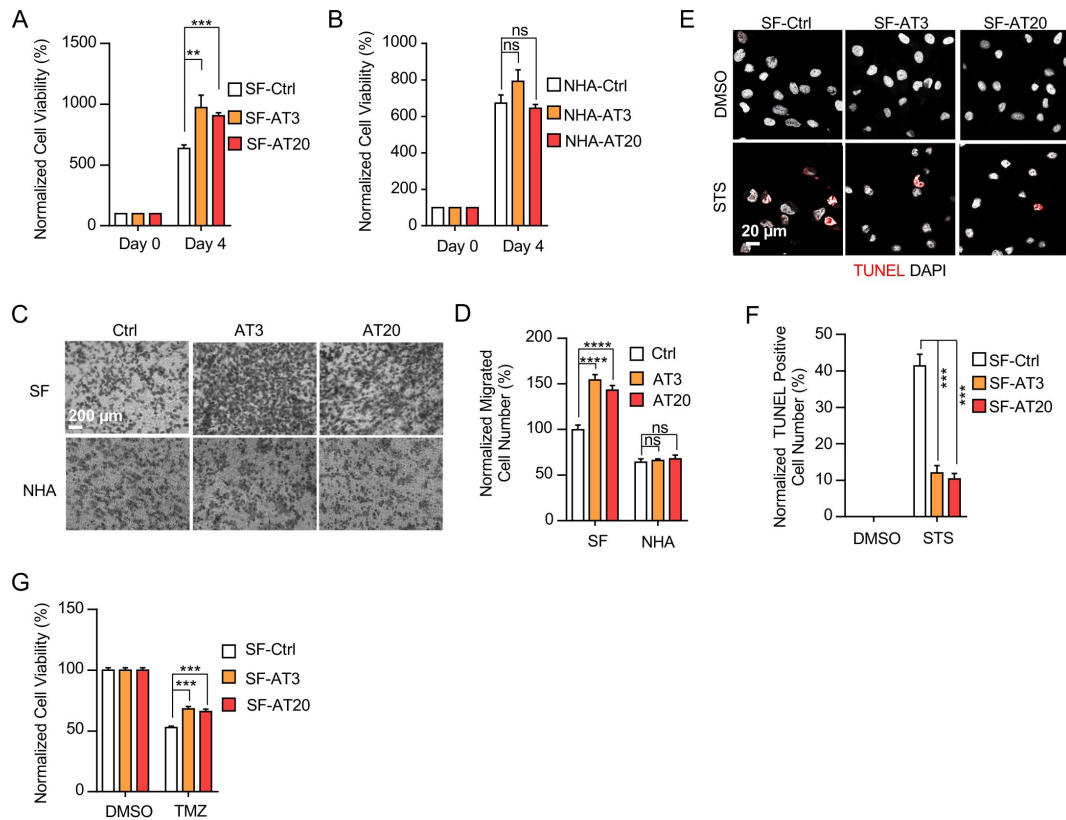


Figure 4.

msiCAT-tailed ATP5α protein promotes GBM progression

- A. MTT assay shows that the expression of ATP5α-AT3 and ATP5α-AT20 promotes GBM cell proliferation. $n=3$, unpaired student's t-test. **, $P < 0.01$; ***, $P < 0.001$.
- B. MTT assay shows that the expression of ATP5α-AT3 and ATP5α-AT20 does not promote NHA cell proliferation. $n=3$, unpaired student's t-test. ns, not significant.
- C. Transwell assay shows that the expression of ATP5α-AT3 and ATP5α-AT20 promotes GBM cell (SF) but not control cell (NHA) migration.
- D. Quantification of C showing the number of migrated cells. $n=3$, unpaired student's t-test. ****, $P < 0.0001$; ns, not significant.
- E. TUNEL staining shows that the expression of ATP5α-AT3 and ATP5α-AT20 can block STS (1 μM)-induced apoptosis in GBM cells. TUNEL-Cy3 was used to indicate apoptotic cells. DAPI was used to stain nucleus.
- F. Quantification of E showing the percentage of TUNEL-positive cells in the population. $n=3$, unpaired student's t-test. ****, $P < 0.0001$.
- G. MTT assay shows that the expression of ATP5α-AT3 and ATP5α-AT20 enhances resistance to temozolomide (TMZ, 150 μM). TMZ treated SF-Ctrl group is used as the control. $n=3$, unpaired student's t-test. ***, $P < 0.001$.

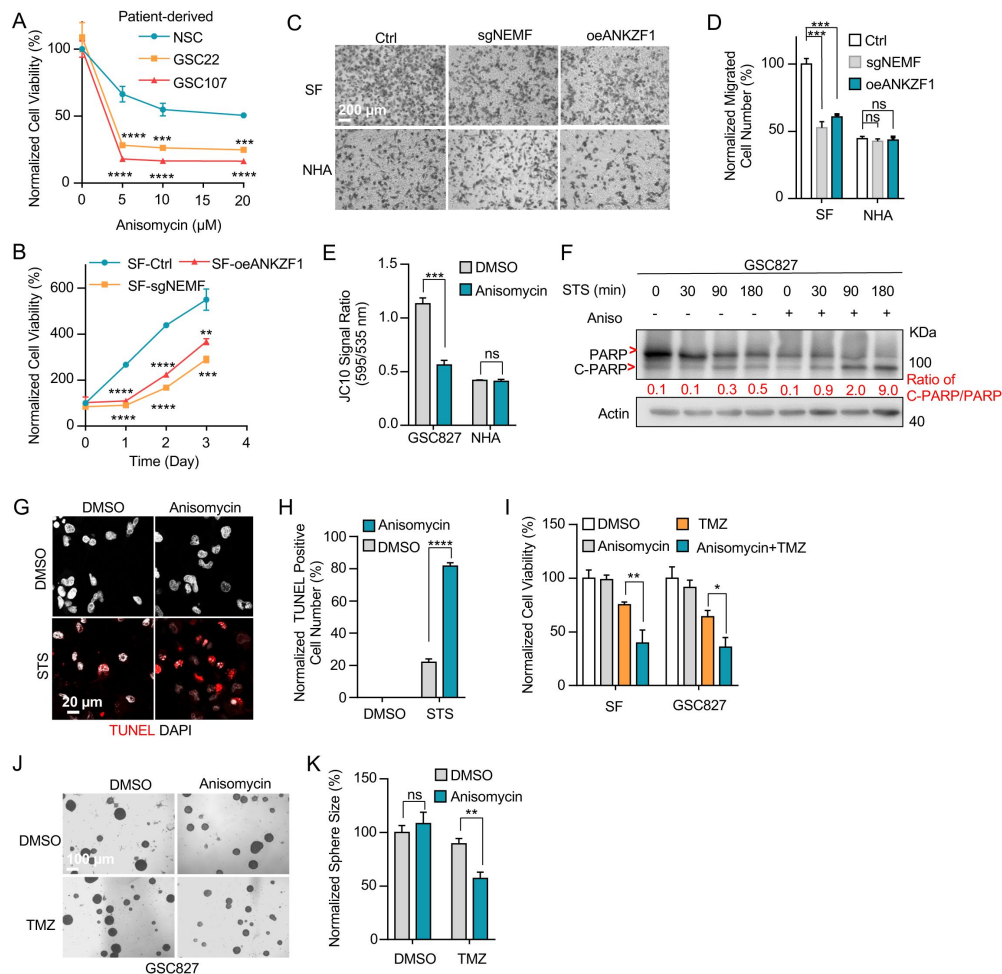


Figure 5.

Inhibition of msiCAT-tailing impedes GBM progression

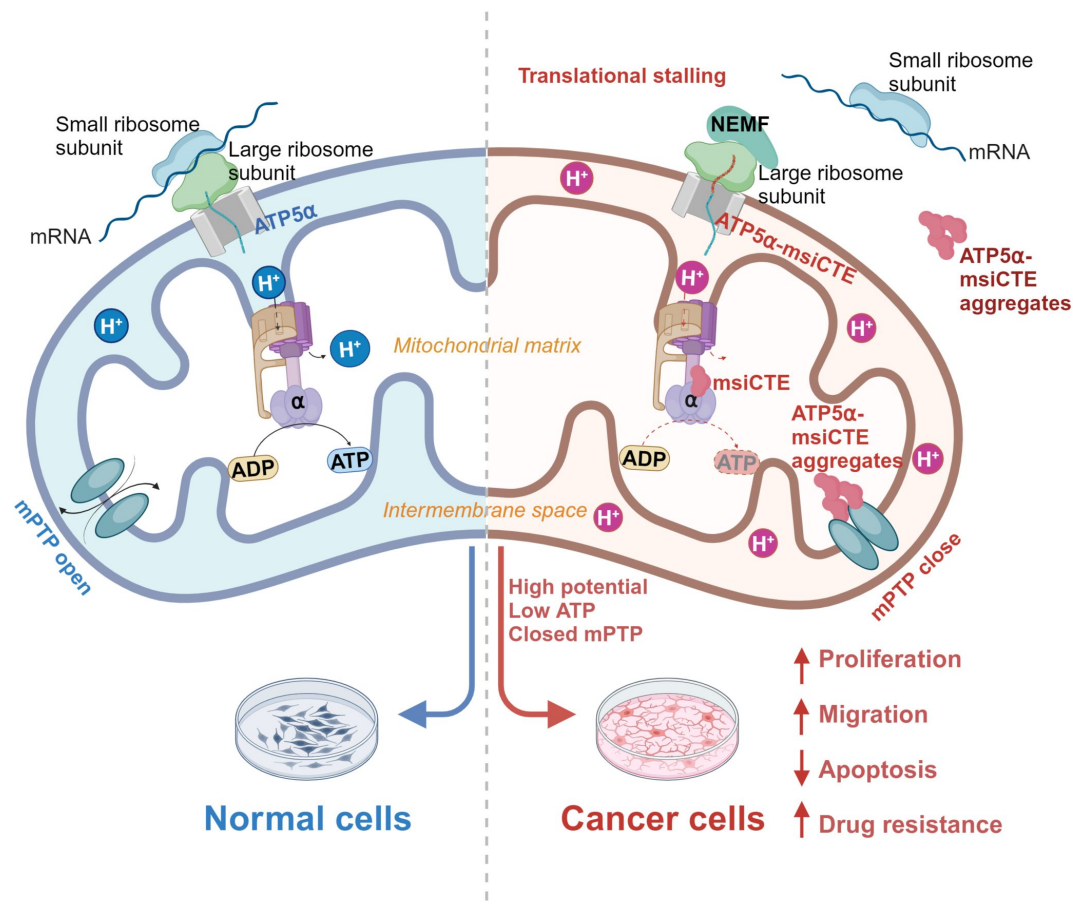
- A. Cell viability assay shows that patient-derived GSC (pGSC) cells are significantly lower than control NSC cells, after 48 hours of anisomycin treatment. ***, $P < 0.001$; ****, $P < 0.0001$, compared to the NSC controls at a certain dose of anisomycin.
- B. Cell viability assay shows that genetic inhibition of msiCAT-tailing pathway hampers GBM cell proliferation. **, $P < 0.01$; ***, $P < 0.001$; ****, $P < 0.0001$, compared to the control group at the corresponding time.
- C. Transwell assay shows that genetic inhibition of msiCAT-tailing pathway blocks GBM cell (SF) but not control cell (NHA) migration.
- D. Quantification of C showing the number of migrated cells. $n=3$, unpaired student's t-test. ***, $P < 0.001$; ns, not significant.
- E. JC-10 staining shows that pharmacological inhibition of msiCAT-tailing synthesis with anisomycin reduces mitochondrial membrane potential in GBM but not in NHA control cells. $n=3$, unpaired student's t-test. ***, $P < 0.001$; ns, not significant.
- F. Western blot analysis of PARP in anisomycin treated and untreated GSC cells shows that pharmacological inhibition of msiCAT-tailing pathway potentiates STS-induced apoptosis. Red numbers below the blots represent the ratio of cleaved PARP (c-PARP) to PARP protein. Actin serves as the loading control.
- G. TUNEL staining shows that pharmacological inhibition of msiCAT-tailing pathway promotes STS (1 μM)-induced apoptosis in GBM cells. TUNEL-Cy3 is used to indicate apoptotic cells. DAPI is used to stain nucleus.
- H. Quantification of G showing the percentage of TUNEL-positive cells in the population. $n=3$, unpaired student's t-test. **** $P < 0.0001$. DMSO is used as the vehicle control.
- I. Cell viability assay shows that pharmacological inhibition of msiCAT-tailing pathway could reduce the resistance of GBM cells to TMZ treatment (150 μM). $n=3$, unpaired student's t-test. * $P < 0.05$; **, $P < 0.01$.
- J. Neurosphere formation assay shows that pharmacological inhibition of msiCAT-tailing pathway can cooperate with TMZ to reduce the formation of spheroids in GBM cells, indicating the reduction of drug resistance.
- K. Quantification of J. $n=3$, unpaired student's t-test. **, $P < 0.01$.

Next, we studied the effect of RQC pathway on the anti-apoptosis of GBM cells. The first thing we discovered were changes in mitochondrial function. Previously, genetic inhibition of the RQC pathway (sgNEMF or oeANKZF1) could significantly reduce the membrane potential of GBM mitochondria (**Fig. 2C**); anisomycin treatment also has the same effect here (**Fig. 5E**). We also found that in GSC cells, anisomycin treatment could promote MPTP opening, but not in NHA cells (**Fig. 3A**). Correspondingly, inhibition of the RQC pathway with anisomycin rendered GBM cell lines more sensitive to STS-induced apoptosis, as evidenced by more PARP-1 cleavage and TUNEL-positive signals (**Fig. 5F–5H**). Similarly, genetic inhibition of the RQC pathway could achieve the similar sensitization (**Fig. 5SD, 5SE**). Finally, we found that the RQC pathway also plays a role in TMZ-induced cell death. For example, co-treatment of anisomycin and TMZ significantly inhibited the survival of GBM cell lines (**Fig. 5I**) and effectively inhibited the size of spheroids formed by GSC cells (**Fig. 5J** and **5K**). In conclusion, the activity of the RQC pathway is important for GBM progression, including their proliferation, migration, and survival upon induction of apoptosis.

Discussion

In this study, we mainly explored the role of RQC and its products, especially the mitochondrial proteins modified by the msiCAT-tailing mechanism, in cancer cells. We used GBM cells as a model to conduct research, but this does not mean that these phenomena only occur in GBMs. They may be widely present in the occurrence and development of cancer. Here, we focused on msiCAT-tail modifications on the mitochondrial protein ATP5 α . In our model, due to the specific function of mitochondrial ATP synthase itself, msiCAT tail-modified proteins endow GBM cells with several unique properties, such as maintaining or increasing mitochondrial membrane potential despite reducing ATP production, promoting the survival and migration of GBM cells, and providing resistance to STS-induced apoptosis, and this last point may be related to regulating the open-close state of MPTP (**Fig. 6**). These characteristics promote the malignancy of tumor cells, thus inhibiting the RQC pathway may be used as an effective adjuvant treatment to complement existing chemotherapy methods. In addition, studying the behavior of ATP synthase in cancer is also of special significance. In carcinogenesis, ATP synthase is often found re-localized to the plasma membrane and is known as the ectopic ATP synthase complex (eATP synthase). These eATP synthases have catalytic activity and can promote the generation of ATP in the extracellular environment to establish a suitable tumor microenvironment (55). eATP synthase was shown to be first assembled in mitochondria and then delivered to the cell surface via microtubules (46). However, what kind of ATP synthase can be delivered to the plasma membrane is not clear. Studying the localization of CAT-tailed eATP synthase in the future may provide us with clues.

It is worth noting that more than one mitochondrial protein can be CAT-tailed in cancer cells. It is very likely that the multiple nuclear-genome encoded mitochondrial proteins have been CAT-tailed in the similar way, and due to the different properties of these base proteins, msiCAT-tailed peptides may have diverse effects on mitochondria or cells. For example, CAT-tailed COX4 protein may contribute to a more significant and direct reduction in mitochondrial respiratory efficacy. It may be interesting to investigate their roles separately, as the combined effects of the individual defects may be key to helping us understand the observed mitochondrial changes in cancer. Indeed, in a recent study, the authors found that knocking down ANKZF1 inhibited the progression of GBM by causing the accumulation of abnormal proteins within mitochondria (29). Combined with our data, we conclude that the balance of ANKZF1 expression and activity is essential for cancer proliferation. Excess or deficiency may cause changes in cell adaptability. A minor drawback is that in their study, the authors used a mitochondrial-localized non-stopped GFP protein to induce proteostasis stress, and there was no direct biochemical evidence for detecting the CAT-tailed proteins. Here, we focus on endogenous proteins to analyze the impact of target proteins on mitochondria in more detail. The consideration is that abundantly expressed non-physiological ectopic proteins may cause general proteostasis failure and thus mask the specific



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Figure 6.

A working model of how msiCAT-tail modified ATP5α protein affects mitochondrial function in GBM cells

In normal cells, the ATP5α protein encoded by the nuclear genome enters the mitochondrial matrix through a co-translational import mechanism through the TOM/TIM complex and is subsequently assembled into ATP synthase (Left Side). In GBM cells, the CAT-tailed ATP5α protein forms insoluble aggregates close to the outer membrane of mitochondria or can be imported into mitochondria. In the mitochondrial matrix, proteins with shorter CAT-tails easily incorporate into ATP synthase and affect its function, which is manifested by a decrease in the ATP synthesis rate and accumulation of protons (H⁺), resulting in an increase in mitochondrial membrane potential. Ultimately, changes in ATP synthase leads to the malfunction of mPTP, thereby affecting cell proliferation, migration and resistance to drug-induced apoptosis (Right Side).

functions of endogenous proteins. In addition, the cell lines used in the two studies were different. GSC is a patient-derived GBM cell line with more stemness, and its mitochondrial status and RQC pathway activity may be different from those of the U87 or U251 cell lines. Therefore, rather than thinking that the conclusions from the two studies are in conflict, it is better to say that they are complementary, and both prove the importance of RQC in tumorigenesis. Our study explores the mechanistic role of RQC pathway in GBM and provides new potential targets for future treatments.

Comprehensive analysis of how many nuclear genome-encoded mitochondrial proteins are modified through the msiCAT-tailing mechanism using advanced mass spectrometry is an interesting topic and deserves further study. In a recent study published in *Cell Reports*, Lv and colleagues found that the cytoplasmic E3 ligase Pirh2 and the mitochondrial protease ClpXP can complement the classic NEMF-ANKZF1 to degrade mitochondrial protein aggregation caused by ribosome stalling (56). The elevation of ClpXP in various cancers (57) may be explained by increased msiCAT-tailing products in mitochondria, and the impact of ClpXP on mitochondrial RQC also depends on future research. ClpXP also regulates the levels of a variety of mitochondrial proteins. In our experiments, we found that ATP5 α proteins without msiCAT-tails are most difficult to ectopically express; short tails (AT3) containing proteins are better, while proteins with long tails (AT20) have higher expression levels and form SDS-insoluble protein aggregates. This regulation may also be achieved through the degradation of ClpXP. Another possibility is regulation at the transcriptional level. The peroxisome proliferator-activated receptor gamma co-activator (PGC-1 α) is a master regulator of mammalian mitochondrial biogenesis (58). PGC-1 α can bind and activate nuclear transcription factors, leading to the transcription of nuclear genome-encoded mitochondrial proteins and the mitochondrial transcription factor Tfam. Tfam subsequently activates transcription and replication of the mitochondrial genome (59). In the future, careful examination of the mRNA levels of the msiCAT-tailed targets and the binding of PGC1 α and Tfam to transcriptional elements will help us distinguish between these two possibilities.

MPTP is an ion nonselective, calcium-dependent, and multifunctional supramolecular channel that penetrates both the inner membranes of mitochondria. Although the function and regulation of MPTP have been widely studied, our understanding of its molecular structure has remained unclear (60). Currently, multiple structural models of MPTP have been proposed: i) the VDAC (voltage-dependent anion channel)/ANT (adenine nucleotide translocator)/Cyp-D (cyclophilin D) model (61). However, subsequent genetic studies have shown that whether these proposed proteins are structural components of MPTP is still highly controversial (62–65). ii) the ATP synthase model of MPTP. In this model, dimers of ATP synthase or reconstitution of ATP synthase (c-ring) may form the molecular structure of MPTP (47, 48). This hypothesis is intriguing and the role of ATP synthase as a component of the pore structure has been the subject of many conflicting studies yet has not yet been unequivocally confirmed. iii) the current prevalent hypothesis is that ANT and ATP synthase together form a large complex (ATP synthasome), and Cyp-D regulates the dynamics of ATP synthasome (66).

MPTP is regulated by the physiological voltage of mitochondrial membrane potential (67, 68), and conversely, the opening of MPTP can also impact ion homeostasis and energy metabolism in the mitochondrial matrix. In our study, two threads converge here. The msiCAT-tailed ATP5 α protein helps cancer cells: i) maintain a high/stable membrane potential, which can desensitize MPTP induction, and ii) directly participate in MPTP assembly, thus inhibiting MPTP's function. Intriguingly, MPTP is an important mediator of cell death. However, for a long time, no reports have been published to clearly confirm or refute the hypothesis, that is more closed (inhibition) MPTP could be an important feature of cancer cells, which can help them escape drug-induced programmed cell death. Here we present evidence supporting this hypothesis. In our observation, GBM cells, especially GSC cells, have a more closed MPTP than control cells. And this state is

directly related to the CAT-tailing modification of the ATP synthase subunit. This is also consistent with evidence that genetic mutations or post-translational modifications in certain ATP synthase subunits can affect MPTP activity (69 [↗](#), 70 [↗](#)).

Materials and methods

Cell lines and cell culture conditions

The human astroglia cell line SVG p12 (ATCC, cat. CRL-8621) and the human glioma cell line SF268 are provided by Dr. Rongze Olivia Lu. Both cell lines were cultured in DMEM (ATCC, cat. #302002) with 10% FBS (Biowest, cat. S1620-100) and penicillin/streptomycin (Gibco™, cat. 15140122). SF268 clones should be maintained in complete DMEM supplemented with 400 µg/mL G418 (Gibco, cat. 10131027). The 0.25% trypsin solution (ATCC, cat. #SM2003C) were used to passage cells. The normal human astrocytes NHA E6/E7/hTERT cell line was from Dr. Russell O. Pieper, UCSF Brain Tumor Research Center. Cells are cultured in ABM™ Basal Medium (Lonza, cat. CC-3187) and AGM™ SingleQuots™ Supplements (Lonza, cat. CC-4123). Corning™ Accutase™ Cell Detachment Solution (Corning, cat. 25058CI) were used to passage cells. GSC827 cells, a patient-derived human glioma stem cell line, was from Dr. Chun-Zhang Yang at NIH. The NSC, NSC26, patient-derived GSC33, GSC22, GSC99, GSC105 and GSC107 cell lines used in this study were kindly provided by Dr. John S Kuo at the University of Texas, Austin. GSC cells were cultured in Neural basal-A Medium (Gibco, cat. #10888022) with 2% B27 (Gibco, cat. #17504044), 1% N2 (Gibco, cat. #17502048), 20 ng/ml of EGF and FGF (Shenandoah Biotechnology Inc. cat. PB-500-017), Antibiotic-Antimycotic (Gibco, cat. #15240062) and L-Glutamine (Gibco, cat. #250300810). Cells could be cultured in both spherical and attached (on Geltrex, Thermo Fisher, cat. A1413202) forms. Corning™ Accutase™ Cell Detachment Solution (Corning, cat. 25058CI) were used to passage cells.

Cells were transfected with X-tremeGENE™ HP DNA Transfection Reagent (Sigma, cat. 6366244001) following the standard protocol. For single clone selection, SF268 cells were treated with 800 µg/ml G418 for 5 days. The cells were then seeded into 96-well plate at a density of 1/100 µL. Positive clones were verified by immunofluorescence staining and immunoblotting. Cells were maintained in complete DMEM containing 400 µg/mL G418.

Primers, Plasmids and viruses

Plasmids pcDNA3.1+/C-(K)-DYK-ATP5F1A (pATP5α control), pcDNA3.1+/C-(K)-DYK-ATP5F1A-AT3 (pATP5α-AT3), pcDNA3.1+/C-(K)-DYK-AT20 (pATP5α-AT20) were generated by GenScript Inc. Plasmids pCMV-5×FLAG-β-globin-control (5FBG-Ctrl) and pCMV-5×FLAG-β-globin-non-stop (5FBG-nonstop) were generated by Dr. Hoshino (Nagoya City University) and Dr. Inada (Tohoku University) (43 [↗](#)).

Viruses (and plasmid used to generate viruses) pLV[CRISPR]-hCas9:T2A:Neo-U6>Scramble[gRNA#1] (sgControl), pLV[CRISPR]-hCas9:T2A:Neo-U6>hNEMF[gRNA#1579] (sgNEMF), pLV[Exp]-Bsd-EF1A>ORF_Stuffer (pLV-control), pLV[Exp]-EGFP:T2A:Puro-EF1A>mCherry (pLV-control-2), pLV[Exp]-Bsd-EF1A>hANKZF1[NM_001042410.2]/HA (oeANKZF1), and pLV[Exp]-mCherry/Neo-EF1A>hANKZF1[NM_001042410.2] (oeANKZF1) were made by VectorBuilder Inc.

Primers (5' to 3') used for RT-PCR are:

<i>HSP60</i>	hsp60_forward: ACCTGCTCTTGAAATTGCC
	hsp60_reverse: CAATCCCTCTTCTCCAAACAC
<i>ACTB</i>	actb_forward: TGTTTGAGACCTTCAACACC
	actb_reverse: ATGTCACGCACGATTTC

Neurosphere formation assay of GSCs

The GSC spheroids were dissociated using Accutase for 2 min. Cells were resuspended in single cell suspension and grown under non-adherent conditions. Cells were seeded in 12-well plates at a density of 0.25×10^6 cells/well and cultured in 3 mL culture medium for 24 hours. 20 nM of anisomycin and 150 μ M of temozolomide (TMZ) were added to the culture medium and treated for 96 hours. Spheroids were images under 10x objective, captured using QCapture, and analyzed with imageJ. Spheroids larger than 50 μ m were counted.

Differential gene expression analysis using the public database

The raw RNA-seq data used to perform the analysis were obtained from the University of California Santa Cruz Xenabrowser (cohort: TCGA TARGET GTEx, dataset ID: TcgaTargetGtex_rsem_gene_tpm, <https://xena.ucsc.edu/>), and then subsets included only TCGA glioma (GBM), GTEx Brain Frontal Cortex, and GTEx Cortex samples. Differential expression analysis was conducted using the “limma” package (R version: 4.3.1). The benjamini-Horchberg method was used for multiple testing correction to control the false discovery rate (FDR). Cut-off of adjusted p-value (adj.P.Val) was set at 0.001; cut-off of the absolute fold change was set at 2 (logFC > 1).

Immunostaining

Cells were cultured on sterile coverslips until 80% confluency. For immunostaining, cells were washed with phosphate buffered saline (PBS) solution thrice. Then, 4% formaldehyde (Thermo Fisher, cat. BP531-500) was applied to cells for fixation for 30 min at room temperature. After fixation, cells were washed with PBS solution containing 0.25% TritonX-100 (PBSTx) (Thermo Fisher, cat. T9284) thrice, and blocked with 5% normal goat serum (Jackson Immuno, cat. 005-000-121) for 1 hour at room temperature. Cells were then incubated with primary antibodies overnight in a humidified chamber at 4°C. The next day, cells were washed by PBSTx thrice and incubated with secondary antibodies for 2 hours at room temperature. After washing, cells were stained with 300 nM DAPI (Thermo Fisher, cat. 57-481-0) for 5 min at room temperature and mounted in Fluoromount-G Anti-Fade solution (Southernbiotech, cat. 0100-35). Images were taken using a Zeiss LSM 800 confocal microscope, 40x oil objective lens and Airyscan processing. The primary antibodies used in the study were rabbit anti-ATP5a (Cell Signaling, cat. #18023), mouse anti-TOMM20 (1:500, Santa Cruz, cat. 18023S), rabbit anti-MTCO2 (1:500, Abclonal, cat. sc-17764). The secondary antibodies were Alexa fluor 633-, 594-, 488-conjugated secondary antibodies (1: 300, Invitrogen, cat. A21071, A11036, A32732).

SDS-PAGE and immunoblotting

Cells or isolated mitochondria were solubilized in cell lysis buffer containing 50mM Tris-HCl pH 7.4, 150 mM NaCl, 10% glycerol, 1% TritonX-100, 5 mM EDTA and 1x protease inhibitor (Bimake, cat. B14002). Protein concentration was measured by using the Bradford assay (BioVision, cat. K813-5000-1). Samples were separated in 4-12% Tris-Glycine gel (Invitrogen, cat. WXP41220BOX) and proteins were transferred to PVDF membrane (Millipore, cat. ISEQ00010). The membranes were then blocked with 5% non-fat dry milk (Kroger) for 50 min at room temperature and probed

with primary antibodies overnight at 4°C. Membranes were washed with Tris buffered saline with 0.1% Tween 20 (TBST) solution thrice and then incubated with secondary antibodies for 1 hour at room temperature. Blots were detected with ECL solution (PerkinElmer, cat. NEL122001EA) and imaged by Chemidoc system (BioRad). The intensity of blots was further analyzed by ImageJ software. The primary antibodies used were mouse anti-Actin (1:1000, Santa Cruz, cat. sc-47778), rabbit anti-NEMF (1:1000, Proteintech, cat. 11840-1-AP), mouse anti-ANKZF1 (1:1000, Santa Cruz, cat. sc-398713), mouse anti-ATP5a (Abcam, cat. ab14748), mouse anti-C-I30 (1:1000, Abcam, cat. ab14711), rabbit anti-COX4 (Abcam, cat. ab209727), mouse anti-Flag (1:1000, Sigma, cat. F1804), rabbit anti-ANT1/2 (1:1000, Proteintech, cat. xxx), rabbit anti-CypD (1:1000, Proteintech, cat. 15997-1-AP). The secondary antibodies used were goat anti-rabbit IgG (1: 5000, Invitrogen, cat. G21234), goat anti-mouse IgG (1:5000, Invitrogen, cat. PI31430).

Mitochondrial isolation, Blue Native PAGE and Western blotting

Cells were homogenized using Dounce homogenizer in ice-cold homogenization buffer containing 210 mM mannitol (Fisher Sci, cat. AA3334236), 70 mM sucrose (Fisher Sci, cat. AA36508A1), 5 mM HEPES (Fisher Sci, cat. 15630106), pH 7.12, 1 mM EGTA (Fisher Sci, cat. 28-071-G), and 1x protease inhibitor. The homogenate was centrifuged at 1500 g for 5 min. The resultant supernatant was centrifuged at 13000 g for 17 min. The supernatant was collected as the cytosol portion and the pellet as mitochondria portion was washed with homogenization buffer and centrifuged at 13000 g for 10 min. For Blue Native PAGE, the mitochondria samples were solubilized in 5 % digitonin (Thermo Fisher, cat. BN2006) on ice for 30 min and then centrifuged at 20000 g for 30 min. The supernatant contains solubilized mitochondrial proteins and mixed with 5% G-250 (GoldBio, cat. C-460-5) and 1x NativePAGE sample buffer (Invitrogen, cat. BN2008) (final G-250 concentration is 25% of the digitonin concentration). Mitochondrial protein concentration was measured by using Bradford assay. Samples were separated in 3-12% Bis-Tris Native gel (Invitrogen, cat. BN1001BOX) and then transferred to PVDF membrane. Membranes were fixed with 8% acetic acid (Thermo Fisher, cat. 9526-33), and then blocked and probed with antibodies as described above for Western blotting.

Mitochondrial membrane potential assays

Mitochondrial membrane potential of GSC cells was measured using Image-iTTM TMRM (Invitrogen, cat. I34361). Cells were cultured in 96-well black plate at a density of 1×10^5 cells per well overnight in incubator with 5% CO₂ at 37°C. Cells were incubated with TMRM (100 nM) for 30 min at 37°C. Then, cells were washed with PBS solution three times. Fluorescence changes at excitation/emission of 548/574 nm were monitored with Cytation 5 plate reader (BioTek). Mitochondrial membrane potential was also measured using JC-10 (AdipoGen, cat. 50-114-6552). Cells were cultured in 96-well black plate at a density of 5×10^4 cells per well overnight in incubator with 5% CO₂ at 37°C. Cells were incubated with JC-10 (10 µg/ml) for 45 min at 37°C. Then, cells were washed with PBS solution twice. Fluorescence changes at excitation/emission of 535/595 nm for JC-10 aggregates and at 485/535 nm for JC-10 monomers were monitored with a Synergy 2 Reader (BioTek). Mitochondrial membrane potential was quantified as the fluorescence of JC-10 aggregates/monomers (595/535 nm).

Oxygen Consumption Rate assay with a Seahorse analyzer

The oxygen consumption rate (OCR) of cells was measured using the Seahorse Cell Mito Stress Test kit following the user guide (Agilent, cat. 103010-100). Briefly, cells were cultured in testing chambers at a density of 8000 cells per well overnight in incubator with 5% CO₂ at 37°C. Cells were then washed with assay medium containing Seahorse XF DMEM medium (Agilent, cat. 103575-100) with 1 mM pyruvate, 2 mM glutamine and 10 mM glucose twice, and incubated in assay medium for 1 hour in incubator without CO₂ at 37°C. Cells were treated with compounds in the order of

oligomycin (1.5 μ M), carbonyl cyanide-4 (trifluoromethoxy), phenylhydrazone (FCCP, 1.0 μ M), and Rotenone/Antimycin (0.5 μ M). The OCR of cells was monitored by using Seahorse XF HS Mini (Agilent).

Mitochondrial MPTP assay

The opening of mitochondrial permeability transition pore was measured using Invitrogen™ Image-IT™ LIVE Mitochondrial Transition Pore Assay Kit (Invitrogen, cat. I35103). Cells were cultured in 35 mm glass-bottom dishes overnight in an incubator with 5% CO₂ at 37°C. Cells were washed twice with the modified Hank's Balanced Salt Solution (HBSS, Thermo Fisher, cat. 14025092) containing 10 mM HEPES, 2 mM L-glutamine and 0.1 mM succinate (Thermo Fisher, cat. 041983.A7) and incubated with the labeling solution (1 μ M Calcein, 0.2 μ M MitoTracker Red, 1 mM Cobalt Chloride) for 15 min at 37°C. Cells were then washed with HBSS twice and imaged at excitation/emission of 494/517 nm for Calcein and at 579/599 nm for MitoTracker Red by using the Zeiss confocal microscope.

Mitochondrial Ca²⁺ retention capacity assay

The mitochondrial calcium retention capacity (CRC) was measured on a Cytation 5 reader at excitation/emission of 506/592 nm using the membrane-impermeable fluorescent probe Calcium green-5N (Invitrogen, cat. C3737). Isolated mitochondria samples (0.75 mg protein/mL) were incubated in 1 mL swelling medium supplemented with 10 mM succinate, 1 μ M Calcium green-5N, and inorganic phosphate and cyclosporine A (Thermo Fisher, cat. AC457970010). One Ca²⁺ addition was 1.25 nmol (1 mL volume). Only the MPTP opening in the presence of cyclosporine A was induced by high amounts of added calcium (30 nmol Ca²⁺ in the last two additions). The CRC value was calculated as total Ca²⁺ accumulated in the mitochondria per unit (1 mg protein).

MTT assay

Cell proliferation was measured by using the MTT assay kit (Roche, cat. 11465007001). Cells were cultured in 96-well plates at a density of 2000 cells per well overnight in an incubator with 5% CO₂ at 37°C. Cells were treated with MTT labeling reagent for 4 hours at 37°C. The solubilization buffer was added to cells and then cells were incubated overnight at 37°C. Absorbance changes of the samples at 550 nm were monitored by using a Synergy 2 Reader (BioTek).

Wound healing assay

Cells were seeded into 6-well plates and cultured for 24-48 hours to reach a confluent cell monolayer. Cells were treated with serum-free medium overnight before mechanical scratching (53 [\[1\]](#)). Images of the wounds were taken at 0, 24 and 48 hours. Wounds areas were measured by using the wound healing plugin of ImageJ. Wound coverage % = 100% × [A_{t=0h} - A_{t=Δh}] / A_{t=0h} (A_{t=0h} is the area of the wound measured immediately after scratching $t = 0h$, A_{t=Δh} is the area of the wound measured h hours after scratch is performed).

Cell migration assay

Cell migration was measured by using Transwell assays (Corning, cat. CLS3422). Cells were cultured in Transwell inserts at a density of 1×10^5 cells per well for 3 hours in an incubator at 37°C with 5% CO₂. The top inserts were supplemented with DMEM medium only, and the bottom wells were supplemented with DMEM medium with 20% Fetal Bovine Serum. After incubation, the cells on the apical side of the Transwell insert membrane were removed using a cotton applicator. The cells on the bottom side of the insert were rinsed with PBS twice and fixed in 70% ethanol (Thermo Fisher, cat. R40135) for 15 min at room temperature. After fixation, inserts were placed into an empty well to allow the membrane to dry. Then, the insert was incubated with 0.2% crystal

violet (Sigma, cat. V5265) for 5 min at room temperature. The insert was rinsed with water twice and images were captured by using a microscope with a 20x objective. Cell numbers were quantified using ImageJ.

TUNEL staining

The apoptosis was measured by a TUNEL assay kit (ApexBio, cat. K1134). Cells were cultured on sterile cover slips until 80% confluency and washed with PBS thrice. Then, 4% formaldehyde was applied to cells and fixed for at 4°C 25 min. After fixation, cells were washed with PBS twice and incubated with 20 μ M proteinase K (Invitrogen, cat. 25530049) for 5 min at room temperature. Then, cells were rinsed with PBS thrice and incubated in 1x equilibration buffer for 10 min at room temperature. Cells were stained with FITC or Cy3 labeling mix for 1 hour at 37°C in a humidified chamber. Cells were washed by PBS thrice and stained with DAPI for 5 min at room temperature. Cells were mounted in Fluoromount-G Anti-Fade solution and imaged at 520 nm for FITC or at 570 nm for Cy3 by using the Zeiss confocal microscope.

Mitochondria ATP measurement via fluorescence imaging of ATP-red

BioTracker™ ATP-red dye (Millipore, cat. SCT045) is a fluorogenic indicator for ATP in mitochondria (71 [↗](#)). Cells cultured in monolayer condition were incubated in medium with 5 μ M ATP-red for 15 min in an incubator at 37°C with 5% CO₂. Mitochondria were also labeled with incubating cells with 100 nM MitoTracker-Green (Invitrogen, cat. M7514) for 15 min to normalize their mass. Before measurement, cells were washed twice with culture medium and then fresh medium was added. Cells were imaged in a 37°C chamber with 5% CO₂ at excitation/emission of 510/570 nm for ATP-red and at excitation/emission of 490/516 nm for MitoTracker-Green by using the Zeiss confocal microscope. The ATP-red signals could be also measured by a Synergy 2 Reader (BioTek).

Co-immunoprecipitation

Cells were lysed in the buffer containing 50 mM Tris-HCl pH 7.4, 150 mM NaCl, 10% glycerol, 1% TritonX-100, 5 mM EDTA and 1x protease inhibitor. Soluble samples were incubated with 1.5 μ L ATP5 α antibody at 4°C with mixing overnight. 25 μ L of protein A/G magnetic beads (Pierce, cat. 88802) were added to the co-IP samples and incubated at 4°C with mixing overnight. Samples were washed with washing buffer thrice and then applied to SDS-PAGE analysis.

Statistics

Statistical analyses were performed by using Graphpad 7.0 software. Chi-squared test and unpaired student's t-test were used for comparison. $P < 0.05$ was considered significant, except in gene expression analysis (Fig. 1A [↗](#)). *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$; ****, $P < 0.0001$; ns, not significant. All data were expressed as means $\pm$ s.e.m.

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Author contributions

TC and BZ designed, performed the experiments, analyzed the data and wrote the manuscript. TC, BZ, ER, YW, YT, IM, and JW performed the experiments and analyzed the data. RL and ZW formed the idea, supervised the project, wrote the manuscript and provided funding.

Conflict of interest

The authors declare no competing interests.

Materials availability

Plasmids and other materials generated in this study will be available upon request from the lead contact with a completed Materials Transfer Agreement.

Supplementary Figure and Table Legends

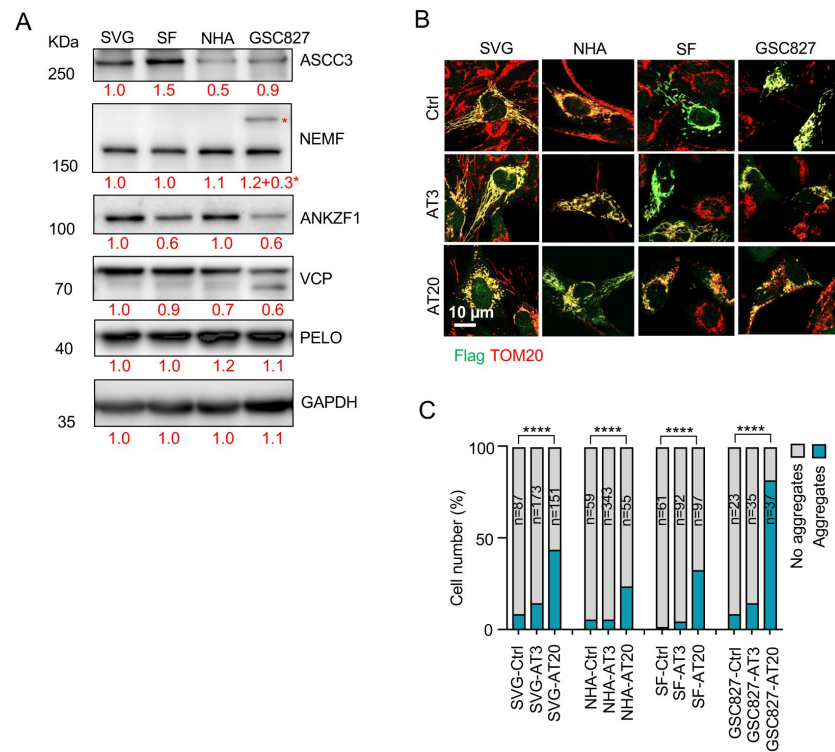


Figure S1.

The RQC pathway is disturbed in GBM cells

A. Western blot analysis of some RQC factors in control (SVG, NHA) and GBM (SF268, GSC827) cell lines. Actin serves as the loading control. Red * indicates a modified form of NEMF in GSC827 cells. Red numbers represent fold changes in protein levels compared to controls (here SVG), labeling of fold changes in the figures below follows the same rule.

B. Immunofluorescence staining shows that Flag-tagged ATP5α-AT3 and ATP5α-AT20 (green) form aggregates in GBM and control cells. TOM20 (red) serves as the mitochondrial marker. In this and subsequent figures, the scale bar is indicated in the images.

C. Quantification of A, n=3, chi-squared test. ****, $P < 0.0001$. The total number of cells counted is indicated in each column.

Table S1.

Differential expression analysis for RQC genes in GBM patients and healthy controls

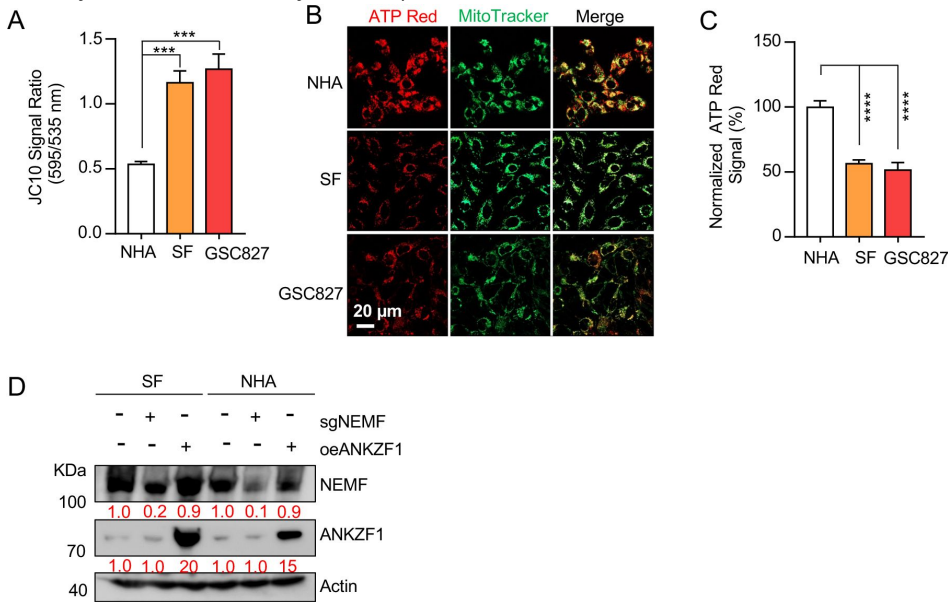
Gene	logFC	AveExpr	t	P.Value	adj.P.Val
RACK1	2.224548565	9.048113333	19.24641934	2.62E-57	6.69E-56
ASCC3	1.738216567	2.717246389	19.27211713	2.05E-57	5.27E-56
ASCC1	1.689584768	4.257915556	17.8774401	1.20E-51	2.18E-50
ASCC2	1.471467207	4.153399167	15.9118075	1.43E-43	1.68E-42
ABCE1	1.32826428	3.81695	13.79248369	4.69E-35	3.60E-34
VCP	1.050066326	6.321021667	9.828944092	2.33E-20	9.31E-20
GIGYF2	0.985695112	3.786421389	10.25440005	8.00E-22	3.41E-21
MAP3K20	0.962218073	1.863793889	8.711292467	1.10E-16	3.75E-16
PELO	0.92860628	2.2885625	10.60122263	4.85E-23	2.18E-22
KLHDC10	0.854921284	3.492322222	9.064976509	8.08E-18	2.88E-17
EDF1	0.82091202	8.444505	7.278600017	2.12E-12	5.94E-12
XRN1	0.809119864	1.518371111	8.54020086	3.80E-16	1.26E-15
LTN1	0.786409776	1.9716	9.962815742	8.14E-21	3.32E-20
MKRN1	0.769369764	5.745359444	7.400071791	9.65E-13	2.74E-12
RCHY1	0.652647968	4.276126111	6.840213538	3.40E-11	8.93E-11
ZNF598	0.62380412	4.006663611	6.043582081	3.76E-09	8.90E-09
HBS1L	0.291107388	4.701389722	2.72853772	0.006672805	0.010370549
NEMF	0.194631373	4.566962778	2.575063266	0.010419695	0.015855894
ANKZF1	-0.436070986	4.620298333	-3.65005718	0.000300886	0.000525859

Abbreviations: RQC, ribosome-associated quality control; GBM, glioblastoma.

Figure S2.

GBM cells have high mitochondrial membrane potentials but low mitochondrial ATP levels

- A. JC-10 staining shows that GBM cells have high mitochondrial membrane potentials than NHA (control) cells. n=3, unpaired student's t-test. ***, $P < 0.001$
- B. BioTracker ATP-red dye staining shows that GBM cells have low mitochondrial ATP production than NHA (control) cells. MitoTracker Green is used to indicate mitochondrial mass.
- C. Quantification of B. n=3, unpaired student's t-test. ****, $P < 0.0001$.
- D. Western blot analysis shows the efficiency of overexpression and knockdown in cells. Actin serves as the loading control.



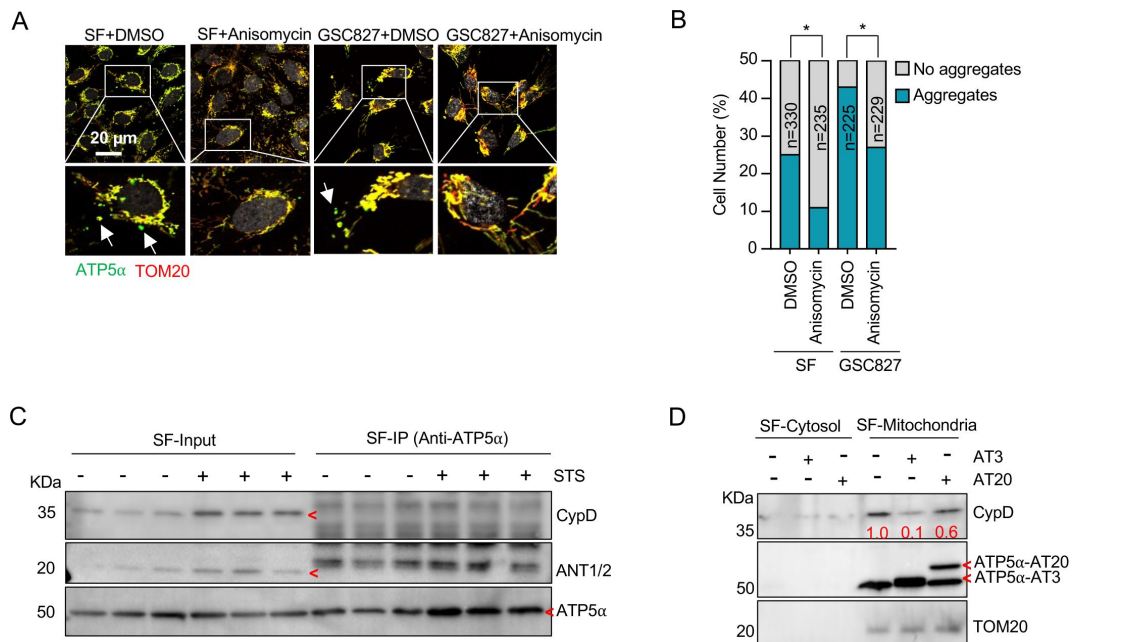


Figure S3.

The regulation of mPTP by CAT-tailing on ATP5 α

A. Immunofluorescence staining shows that anisomycin (200 nM) treatment inhibits protein aggregation formed by endogenous ATP5 α . TOM20 (red) serves as the mitochondrial marker.

B. Quantification of A, $n=3$, chi-squared test. *, $P < 0.05$. The total number of cells counted is indicated in each column.

C. Co-immunoprecipitation shows no direct interaction between ATP5 α and CypD or ANT1/2 can be found in GBM cells. Red arrowheads indicate target proteins.

D. Western blotting of cytosolic and isolated mitochondrial samples shows that expression of ATP5 α -AT3 reduces CypD levels in GBM cells. TOM20 serves as the mitochondrial marker and loading control.

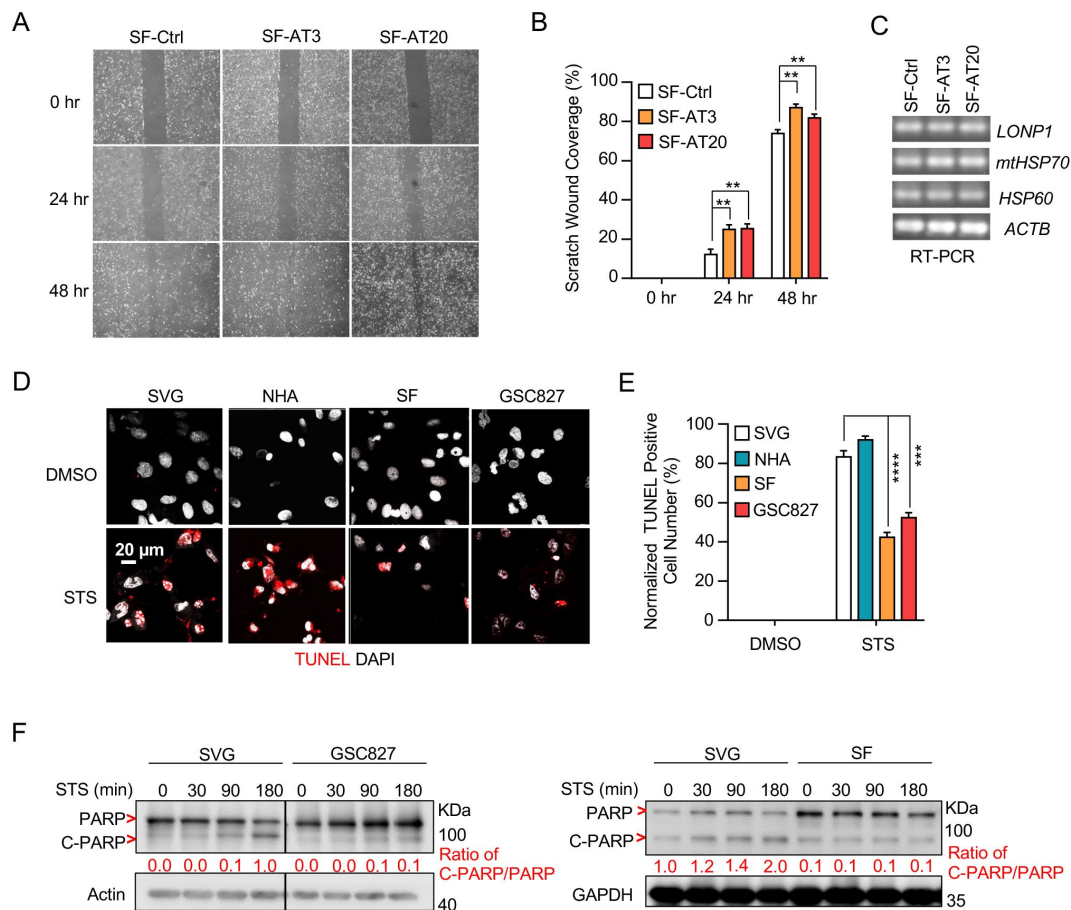


Figure S4.

msiCAT-tailing regulates cell survival in GBM cells

A. Wound healing assay shows that the expression of ATP5 α -AT3 and ATP5 α -AT20 promotes GBM cell (SF268) migration.

B. Quantification of A showing the healing rate indicated by scratch wound coverage after 24 and 48 hours. n=3, unpaired student's t-test. **, $P < 0.01$.

C. RT-PCR shows no significant increase of mitochondrial unfolded protein response markers. Actin serves as the loading control.

D. TUNEL staining shows that GBM cells are more resistant to STS (1 μ M)-induced apoptosis than control cells. TUNEL-Cy3 is used to indicate apoptotic cells. DAPI is used to stain nucleus.

E. Quantification of C showing the percentage of TUNEL-positive cells in the population. n=3, unpaired student's t-test. ***, $P < 0.0001$; ****, $P < 0.0001$.

F. Western blot analysis of PARP shows that GBM cells are more resistant to apoptosis induced by 30, 90, 180 min of STS (1 μ M) treatment. Cleaved PARP serves as an indicator for apoptosis. Actin and GAPDH are used as loading controls. Red numbers below the blots represent the ratio of cleaved PARP (c-PARP) to PARP protein.

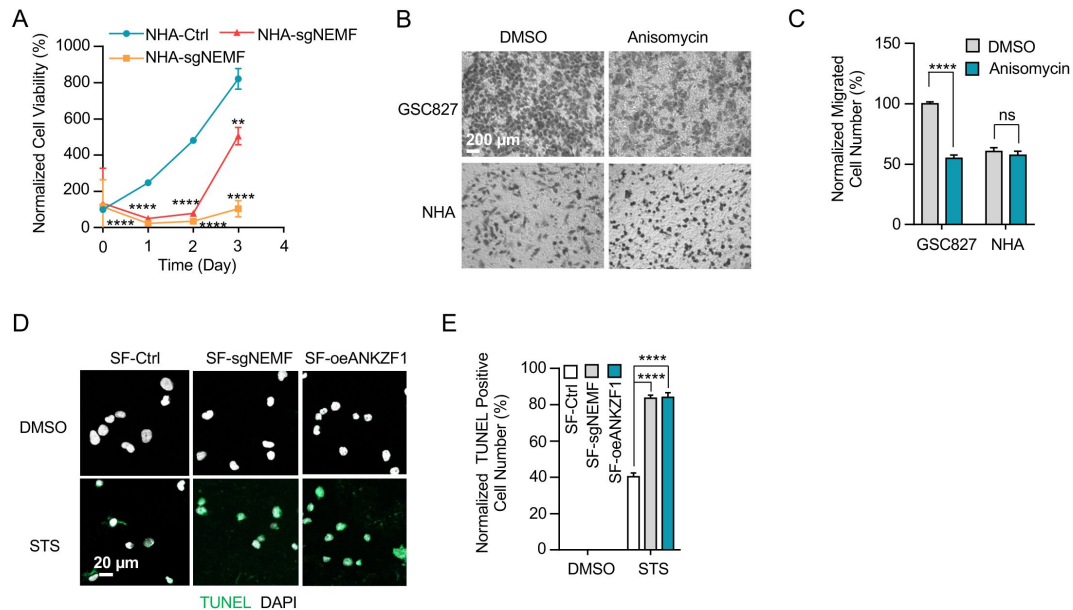


Figure S5.

Inhibition of msiCAT-tailing promotes apoptosis

A. Cell viability assay shows that genetic inhibition of msiCAT-tailing pathway also hampers NHA cell proliferation. **, $P < 0.01$; ****, $P < 0.0001$, compared to the control group at the corresponding time.

B. Transwell assay shows that pharmacological inhibition of msiCAT-tailing pathway blocks GSC cell (SF) but not control cell (NHA) migration.

C. Quantification of C showing the number of migrated cells. $n=3$, unpaired student's t-test. ****, $P < 0.0001$; ns, not significant.

D. TUNEL staining shows that genetic inhibition of msiCAT-tailing pathway promotes STS ($1 \mu\text{M}$)-induced apoptosis in GBM cells. TUNEL-FITC is used to indicate apoptotic cells. DAPI is used to stain nucleus.

E. Quantification of G showing the percentage of TUNEL-positive cells in the population. $n=3$, unpaired student's t-test. **** $P < 0.0001$. DMSO is used as the vehicle control.

References

1. Robichaud N, et al. (2019) **Translational Control in Cancer** *Cold Spring Harbor perspectives in biology* **11**
2. Dever TE, Green R (2012) **The elongation, termination, and recycling phases of translation in eukaryotes** *Cold Spring Harbor perspectives in biology* **4**
3. Kim KQ, Zaher HS (2022) **Canary in a coal mine: collided ribosomes as sensors of cellular conditions** *Trends in biochemical sciences* **47**
4. Brandman O, et al. (2012) **A ribosome-bound quality control complex triggers degradation of nascent peptides and signals translation stress** *Cell* **151**
5. Juszkievicz S, Hegde RS (2017) **Initiation of Quality Control during Poly(A) Translation Requires Site-Specific Ribosome Ubiquitination** *Molecular cell* **65**
6. Sundaramoorthy E, et al. (2017) **ZNF598 and RACK1 Regulate Mammalian Ribosome-Associated Quality Control Function by Mediating Regulatory 40S Ribosomal Ubiquitylation** *Molecular cell* **65**
7. Hashimoto S, et al. (2020) **Identification of a novel trigger complex that facilitates ribosome-associated quality control in mammalian cells** *Scientific reports* **10**
8. Juszkievicz S, et al. (2020) **The ASC-1 Complex Disassembles Collided Ribosomes** *Molecular cell* **79**
9. Shao S, Hegde RS (2014) **Reconstitution of a minimal ribosome-associated ubiquitination pathway with purified factors** *Molecular cell* **55**
10. Shen PS, et al. (2015) **Protein synthesis. Rqc2p and 60S ribosomal subunits mediate mRNA-independent elongation of nascent chains** *Science (New York, NY)* **347**
11. Verma R, et al. (2018) **Vms1 and ANKZF1 peptidyl-tRNA hydrolases release nascent chains from stalled ribosomes** *Nature* **557**
12. Kostova KK, et al. (2017) **CAT-tailing as a fail-safe mechanism for efficient degradation of stalled nascent polypeptides.** *Science (New York NY)* **357**
13. Lytvynenko I, et al. (2019) **Alanine Tails Signal Proteolysis in Bacterial Ribosome-Associated Quality Control** *Cell* **178**
14. Sitron CS, et al. (2020) **Aggregation of CAT tails blocks their degradation and causes proteotoxicity in *S. cerevisiae*** *PloS one* **15**
15. Choe YJ, et al. (2016) **Failure of RQC machinery causes protein aggregation and proteotoxic stress** *Nature* **531**
16. Yonashiro R, et al. (2016) **The Rqc2/Tae2 subunit of the ribosome-associated quality control (RQC) complex marks ribosome-stalled nascent polypeptide chains for aggregation** *eLife* **5**

17. Wu Z, et al. (2019) **MISTERMINATE Mechanistically Links Mitochondrial Dysfunction With Proteostasis Failure** *Molecular cell* **75**
18. Li S, et al. (2020) **Quality-control mechanisms targeting translationally stalled and C-terminally extended poly(GR) associated with ALS/FTD** *Proc Natl Acad Sci U S A* **117**:25104–15
19. Rimal S, et al. (2021) **Inefficient quality control of ribosome stalling during APP synthesis generates CAT-tailed species that precipitate hallmarks of Alzheimer’s disease** *Acta neuropathologica communications* **9**
20. Wang N, Wang D (2021) **Genome-wide transcriptome and translome analyses reveal the role of protein extension and domestication in liver cancer oncogenesis** *Molecular genetics and genomics : MGG* **296**
21. Champagne J, et al. (2021) **Oncogene-dependent sloppiness in mRNA translation** *Molecular cell* **81**
22. Rubio A, et al. (2021) **Ribosome profiling reveals ribosome stalling on tryptophan codons and ribosome queuing upon oxidative stress in fission yeast** *Nucleic acids research* **49**
23. Dango S, et al. (2011) **DNA unwinding by ASCC3 helicase is coupled to ALKBH3-dependent DNA alkylation repair and cancer cell proliferation** *Molecular cell* **44**
24. Gao J, et al. (2020) **Suppression of ABCE1-Mediated mRNA Translation Limits N-MYC-Driven Cancer Progression** *Cancer research* **80**
25. Zhou X, et al. (2019) **High ANKZF1 expression is associated with poor overall survival and recurrence-free survival in colon cancer. Future oncology (London England)** **15**
26. Costantini S, et al. (2021) **Valosin-Containing Protein (VCP)/p97: A Prognostic Biomarker and Therapeutic Target in Cancer** *International journal of molecular sciences* **22**
27. Bi X, et al. (2005) **Drosophila caliban, a nuclear export mediator, can function as a tumor suppressor in human lung cancer cells** *Oncogene* **24**
28. Yang Q, Gupta R (2017) **Zinc finger protein 598 inhibits cell survival by promoting UV-induced apoptosis** *Oncotarget* **9**
29. Li G, et al. (2024) **ANKZF1 knockdown inhibits glioblastoma progression by promoting intramitochondrial protein aggregation through mitoRQC** *Cancer letters*
30. Gehrke S, et al. (2015) **PINK1 and Parkin control localized translation of respiratory chain component mRNAs on mitochondria outer membrane** *Cell metabolism* **21**
31. Forrest MD (2015) **Why cancer cells have a more hyperpolarised mitochondrial membrane potential and emergent prospects for therapy** *bioRxiv*
32. Shi Y, et al. (2019) **Gboxin is an oxidative phosphorylation inhibitor that targets glioblastoma** *Nature* **567**
33. Guièze R, et al. (2019) **Mitochondrial Reprogramming Underlies Resistance to BCL-2 Inhibition in Lymphoid Malignancies** *Cancer cell* **36**

34. Ramamoorthy MD, et al. (2018) **Reserpine Induces Apoptosis and Cell Cycle Arrest in Hormone Independent Prostate Cancer Cells through Mitochondrial Membrane Potential Failure** *Anti-cancer agents in medicinal chemistry* **18**
35. Heerdt BG, et al. (2006) **Growth properties of colonic tumor cells are a function of the intrinsic mitochondrial membrane potential** *Cancer research* **66**
36. Zorova LD, et al. (2018) **Mitochondrial membrane potential** *Analytical biochemistry* **552**
37. Maria A, et al. (2020) **Biochemistry, Electron Transport Chain**
38. Liberti MV, Locasale JW (2016) **The Warburg Effect: How Does it Benefit Cancer Cells?** *Trends in biochemical sciences* **41**
39. Varn FS, et al. (2022) **Glioma progression is shaped by genetic evolution and microenvironment interactions** *Cell* **185**
40. Rutka JT, et al. (1987) **Establishment and characterization of five cell lines derived from human malignant gliomas** *Acta neuropathologica* **75**
41. Kim E, et al. (2013) **Phosphorylation of EZH2 activates STAT3 signaling via STAT3 methylation and promotes tumorigenicity of glioblastoma stem-like cells** *Cancer cell* **23**
42. Sonoda Y, et al. (2001) **Formation of intracranial tumors by genetically modified human astrocytes defines four pathways critical in the development of human anaplastic astrocytoma** *Cancer research* **61**
43. Saito S, et al. (2013) **The Hbs1-Dom34 protein complex functions in non-stop mRNA decay in mammalian cells** *The Journal of biological chemistry* **288**
44. Osuna BA, et al. (2017) **In vitro analysis of RQC activities provides insights into the mechanism and function of CAT tailing** *eLife* **6**
45. Morgenstern M, et al. (2017) **Definition of a High-Confidence Mitochondrial Proteome at Quantitative Scale** *Cell reports* **19**
46. Chang YW, et al. (2023) **Spatial and temporal dynamics of ATP synthase from mitochondria toward the cell surface** *Communications biology* **6**
47. Alavian KN, et al. (2014) **An uncoupling channel within the c-subunit ring of the F1FO ATP synthase is the mitochondrial permeability transition pore** *Proceedings of the National Academy of Sciences of the United States of America* **111**
48. Giorgio V, et al. (2013) **Dimers of mitochondrial ATP synthase form the permeability transition pore** *Proceedings of the National Academy of Sciences of the United States of America* **110**
49. Bonora M, et al. (2013) **Role of the c subunit of the FO ATP synthase in mitochondrial permeability transition.** *Cell cycle (Georgetown Tex)* **12**
50. Ichas F, et al. (1997) **Mitochondria are excitable organelles capable of generating and conveying electrical and calcium signals** *Cell* **89**

51. Stockert JC, et al. (2018) **Tetrazolium salts and formazan products in Cell Biology: Viability assessment, fluorescence imaging, and labeling perspectives** *Acta histochemica* **120**
52. Justus CR, et al. (2014) **In vitro cell migration and invasion assays** *Journal of visualized experiments : JoVE* **88**
53. Grada A, et al. (2017) **Research Techniques Made Simple: Analysis of Collective Cell Migration Using the Wound Healing Assay** *The Journal of investigative dermatology* **137**
54. Mashimo M, et al. (2021) **The 89-kDa PARP1 cleavage fragment serves as a cytoplasmic PAR carrier to induce AIF-mediated apoptosis** *The Journal of biological chemistry* **296**
55. Comelli M, et al. (2016) **F1FO ATP Synthase Is Expressed at the Surface of Embryonic Rat Heart-Derived H9c2 Cells and Is Affected by Cardiac-Like Differentiation** *Journal of cellular biochemistry* **117**
56. Lv L, et al. (2024) **NEMF-mediated Listerin-independent mitochondrial translational surveillance by E3 ligase Pirh2 and mitochondrial protease ClpXP** *Cell reports* **43**
57. Cormio A, et al. (2021) **Mitochondrial Caseinolytic Protease P: A Possible Novel Prognostic Marker and Therapeutic Target in Cancer** *International journal of molecular sciences* **22**
58. Ventura-Clapier R, et al. (2008) **Transcriptional control of mitochondrial biogenesis: the central role of PGC-1alpha** *Cardiovascular research* **79**
59. Wu Z, et al. (1999) **Mechanisms controlling mitochondrial biogenesis and respiration through the thermogenic coactivator PGC-1** *Cell* **98**
60. Endlicher R, et al. (2023) **The Mitochondrial Permeability Transition Pore-Current Knowledge of Its Structure, Function, and Regulation, and Optimized Methods for Evaluating Its Functional State** *Cells* **12**
61. Beutner G, et al. (1998) **Complexes between porin, hexokinase, mitochondrial creatine kinase and adenylate translocator display properties of the permeability transition pore. Implication for regulation of permeability transition by the kinases** *Biochimica et biophysica acta* **1368**
62. Baines CP, et al. (2007) **Voltage-dependent anion channels are dispensable for mitochondrial-dependent cell death** *Nature cell biology* **9**
63. Gutiérrez-Aguilar M, et al. (2014) **Genetic manipulation of the cardiac mitochondrial phosphate carrier does not affect permeability transition** *Journal of molecular and cellular cardiology* **72**
64. Kokoszka JE, et al. (2004) **The ADP/ATP translocator is not essential for the mitochondrial permeability transition pore** *Nature* **427**
65. Karch J, et al. (2019) **Inhibition of mitochondrial permeability transition by deletion of the ANT family and CypD** *Science advances* **5**
66. Beutner G, et al. (2017) **The Mitochondrial Permeability Transition Pore and ATP Synthase** *Handbook of experimental pharmacology* **240**

67. Petronilli V, et al. (1994) **Regulation of the permeability transition pore, a voltage-dependent mitochondrial channel inhibited by cyclosporin A** *Biochimica et biophysica acta* **1187**
68. Boyman L, et al. (2019) **Dynamics of the mitochondrial permeability transition pore: Transient and permanent opening events** *Archives of biochemistry and biophysics* **666**
69. Antoniel M, et al. (2018) **The unique histidine in OSCP subunit of F-ATP synthase mediates inhibition of the permeability transition pore by acidic pH** *EMBO reports* **19**
70. Carraro M, et al. (2020) **The Unique Cysteine of F-ATP Synthase OSCP Subunit Participates in Modulation of the Permeability Transition Pore** *Cell reports* **32**
71. Wang L, et al. (2016) **A Multisite-Binding Switchable Fluorescent Probe for Monitoring Mitochondrial ATP Level Fluctuation in Live Cells** *Angewandte Chemie (International ed in English)* **55**

Editors

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Reviewer #1 (Public Review):

Summary:

Cai et al have investigated the role of msiCAT-tailed mitochondrial proteins that frequently exist in glioblastoma stem cells. Overexpression of msiCAT-tailed mitochondrial ATP synthase F1 subunit alpha (ATP5) protein increases the mitochondrial membrane potential and blocks mitochondrial permeability transition pore formation/opening. These changes in mitochondrial properties provide resistance to staurosporine (STS)-induced apoptosis in GBM cells. Therefore, msiCAT-tailing can promote cell survival and migration, while genetic and pharmacological inhibition of msiCAT-tailing can prevent the overgrowth of GBM cells.

Strengths:

The CAT-tailing concept has not been explored in cancer settings. Therefore, the present provides new insights for widening the therapeutic avenue.

Weaknesses:

Although the paper does have strengths in principle, the weaknesses of the paper are that these strengths are not directly demonstrated. The conclusions of this paper are mostly well-supported by data, but some aspects of image acquisition and data analysis need to be clarified and extended.

<https://doi.org/10.7554/eLife.99438.1.sa2>

Reviewer #2 (Public Review):

This work explores the connection between glioblastoma, mito-RQC, and msiCAT-tailing. They build upon previous work concluding that ATP5alpha is CAT-tailed and explore how CAT-tailing may affect cell physiology and sensitivity to chemotherapy. The authors conclude that when ATP5alpha is CAT-tailed, it either incorporates into the proton pump or aggregates and that these events dysregulate MPTP opening and mitochondrial membrane potential and that this regulates drug sensitivity. This work includes several intriguing and novel observations connecting cell physiology, RQC, and drug sensitivity. This is also the first time this reviewer has seen an investigation of how a CAT tail may specifically affect the function of a protein. However, some of the conclusions in this work are not well supported. This significantly weakens the work but can be addressed through further experiments or by weakening the text.

<https://doi.org/10.7554/eLife.99438.1.sa1>

Author response:

We are grateful for the reviewers' acknowledgment of the originality of our manuscript and its potential importance in cancer treatment. We appreciate the reviewers' critiques on certain conclusions and thank them for their thorough feedback on the manuscript. In the revised version, we will provide a more detailed clarification of the previous data and methods, bolster the existing data, and present additional evidence in support of our hypothesis. Please find below our replies to particular concerns.

In brief, to address the comments from Reviewer 1, we will make the following revisions in the manuscript:

- (1) To discuss the issues regarding the specificity of ATP5 α CAT-tailing, we will provide new patient-derived cell lines and tumor samples and investigate the CAT-tail modifications of nuclear genome-encoded mitochondrial proteins and changes in RQC proteins within them. We will endeavor to explore the nature of NEMF modifications in GSC cells (Fig. S1A).
- (2) To enhance the quality of image data, we will substitute some images (such as Fig. 1E and 3A) with higher quality images.
- (3) To further understand the influence of NEMF on cancer, the effects of NEMF overexpression in GSC cells will be evaluated through testing (*e.g.*, Fig. 3D).
- (4) To further explore changes in apoptosis, we will employ additional methods to detect apoptosis, including Annexin-PI FACS assays, caspase cleavage analysis, assessing BAX-BCL2 ratios, and monitoring cytochrome c release.
- (5) To further confirm the effectiveness of the CAT-tailing-mitochondria mechanism in *in vivo* tumor models, we will utilize a *Drosophila* model to study the impact of the RQC pathway and CAT-tailing mechanism on tumor proliferation *in vivo*. The overactivation of the Notch signaling pathway in *Drosophila* can stimulate malignant proliferation of neural stem cells (NSCs) through both canonical (c-Myc mediated pathway) and non-canonical (PINK1-mitochondrial-mTORC2 pathway) pathways, leading to the development of a tumor-like phenotype in the larval brain. A recent publication in PNAS Nexus (Khaket et al., *PNAS Nexus*, 2024) discusses the impact of the RQC pathway on c-Myc. It is possible for us to analyze the alterations in CAT-tailing on mitochondrial proteins and mitochondrial membrane potential in this Notch model and study how the RQC pathway regulates them. Moreover, tumor implantation experiments will be carried out using immunodeficient mice. Our goal is to conduct a comparative analysis of the growth of control and NEMF KD glioblastoma cell lines in animal models, alongside performing essential biochemical analyses.

Reference:

Khaket, T. P., et al. (2024). Ribosome stalling during c-myc translation presents actionable cancer cell vulnerability. *PNAS nexus*, 3(8), pgae321.

To address the comments from Reviewer 2, we will make the following revisions in the manuscript:

(1) The concerns raised by the reviewer regarding the authenticity of the ATP5a CAT-tail modification are duly noted. Critical control experiments will be incorporated into our study, including *NEMF* knockout (or NFACT domain mutants) and cycloheximide treatment, alongside other methodologies. The results of these experiments will include placements such as Fig. 1B, 1C, S3A, and S3B to improve comprehension of the CAT-tail modification on ATP5 α .

(2) We thank the reviewer for reminding us to consider the differences between the artificial tail and the endogenous CAT-tail. A recently published study (Khan et al., 2024) provides a thorough analysis of the components of the CAT-tail. Our approach to addressing this issue involves emphasizing the use of the artificial CAT-tail sequence and adopting a more measured tone in the revised version. Additionally, we will induce the endogenous ATP5 α -CAT-tail by express *ATP5 α -K20-non-stop* in cells to validate their function in glioblastoma cells.

(3) Moreover, we aim to examine the impact of different amino acid compositions in the ATP5 α c-terminus extension, such as the poly (Gly-Ser) repeats noted by the reviewer, on both mitochondrial function and glioblastoma biology in our revision. By comparing the results obtained from ATP5 α -CAT-tails with different compositions, it is anticipated that more definitive conclusions can be drawn.

(4) Additional minor revisions will be implemented to the text in accordance with the feedback given by the reviewer.

Reference:

Khan, D., Vinayak, A. A., Sitron, C. S., & Brandman, O. (2024). Mechanochemical forces regulate the composition and function of CAT tails. *bioRxiv*, 2024-08.

<https://doi.org/10.7554/eLife.99438.1.sa0>